

Causal-Order Advantage Estimation: Provenance-Aware Credit Assignment for Reinforcement Learning in Dynamic Task Assignment

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Abstract

Dynamic task assignment consists of routing arriving work to a limited number of resources. When such a problem is modelled as an Action-Evolution Petri Net (A-E PN) and solved with deep reinforcement learning, the simulator records the *provenance* of every token, that is, which earlier decisions each reward descends from, and existing learning algorithms do not use this record. This paper proposes to use it for credit assignment. Generalized Advantage Estimation accumulates temporal-difference errors along the trajectory index, but in a system running many concurrent cases the next decision usually belongs to a different case, so its error carries little information about the current action. We run the same recursion along the provenance graph instead. The resulting estimator enters each reward once, at the latest decision on its provenance, lets earlier decisions receive it through the recursion, and combines provenance successors by a flow-weighted average normalized over successors; we show that the alternative normalization, over predecessors, leaves the bootstrap coefficient unbounded and produces an error that grows with the critic. Under a component mean-exogeneity assumption, whose scope we discuss, scoping credit to a decision’s provenance descendants leaves the expected policy gradient unchanged and reduces its variance by a factor governed by the number of reward-bearing causal components, K . On six problem configurations spanning $K = 1$ to $K = 8$, the method improves normalized return over discount-matched PPO by +0.19 to +0.55 (paired, 20 seeds, $p \leq 0.002$) at the four largest K and is indistinguishable from it at the two smallest, at a cost of about one millisecond per episode. Because K is computable from untrained rollouts, it indicates in advance where the method can be expected to help. All experiments are conducted within the A-E PN ecosystem.

Keywords: reinforcement learning, credit assignment, dynamic task assignment, Petri nets, decomposition, policy gradient

1. Introduction

Dynamic task assignment is the problem of deciding, repeatedly and under uncertainty, which resource should serve which unit of work. It covers machine scheduling, service operations, patient flow, order picking and vehicle dispatching. What makes the modern instances hard is the combination of stochastic arrivals, contended and heterogeneous resources, and an action space whose size changes with the state. Recent work models such problems as

Action-Evolution Petri Nets (A-E PN) and solves them with deep reinforcement learning (RL), which accommodates exactly that kind of action space (Lo Bianco et al., 2023, 2025b,a).

In that line of work the model and the learner are developed separately. The net is compiled into a semi-Markov decision process and handed to Proximal Policy Optimization (PPO) with Generalized Advantage Estimation (GAE), and the learning algorithm sees only states, actions and rewards. The execution trace that the simulator produces while generating those quantities is not used as a source of learning signal. This paper asks what is gained by using it.

The difficulty we address is credit assignment in concurrent systems. GAE accumulates temporal-difference errors along the trajectory index, $A_t = \delta_t + \gamma \lambda A_{t+1}$. In an A-E PN running many cases at once, decision $t+1$ typically belongs to a different case from decision t , so its reward is generated by tokens that the action at t never touched. Credit therefore propagates along wall-clock order, while the dependence between decisions and rewards runs along the flow of tokens. This mismatch is a consequence of concurrency, not of the state encoding, and the variance it contributes grows with the number of cases in flight.

The simulator already records what is needed to remove the mismatch. For every transition firing, an A-E PN simulation logs which tokens were consumed, which were produced, when the firing occurred and what reward it emitted. Chaining these records by token identity yields the *provenance* of every reward: the set of earlier decisions from whose outputs the reward’s consumed tokens descend. Provenance is a by-product of executing the net; it is not estimated from data and it is not, by itself, a causal effect. It gives an exact *ancestry* relation, a necessary condition for one decision to have influenced one reward, and we use it to decide which rewards a decision’s advantage is allowed to see. We keep this distinction explicit throughout (Section 3.2).

We leave the policy-gradient algorithm, the network and the objective untouched and change one thing: the successor relation along which the advantage recursion runs. Instead of accumulating from decision t to decision $t+1$ in firing order, we accumulate from a decision to its *provenance successors*, the decisions that consume tokens descending from it. Each reward enters the recursion once, at a single owner, the latest decision on its provenance; earlier decisions on that provenance receive it through the recursion, as in GAE. Ownership therefore partitions the episode’s reward and the estimator is a return decomposition. Successors are combined by a weighted average of their token-flow shares. The construction is purely structural: it needs no token-value semantics, no hand-specified factorization and no re-run simulation. We call it causal-order advantage estimation, where *causal order* refers to the provenance ancestry just described.

The contributions of this paper are the following.

1. A provenance-aware advantage estimator, *CGAE-CF* (Section 4): GAE with the temporal successor replaced by the provenance successor, single-owner reward attribution and a convex flow-weighted combination over successors. The credit factorization is derived from the simulator’s record, per episode, at a cost of about one millisecond.
2. The conditions under which the construction is justified, and a bias analysis (Section 5). We state a component

mean-exogeneity assumption, discuss when operational systems satisfy it, isolate a structural condition on reward firings that is checkable on the trace, prove that under both the closure estimand is unbiased for the policy gradient, and bound the departure of the shipped estimator from that estimand by a quantity computable before training.

3. The identification and repair of a normalization defect. The natural flow-weighted form normalizes over the predecessors of each node, which is what token conservation yields; but the recursion sums over successors, so the coefficient multiplying the bootstrap is a row sum that the constraint does not bound, and the resulting error is proportional to the critic. Normalizing over successors removes it identically.
4. A pre-training diagnostic. The number of reward-bearing causal components K and the mean causal fan-out are computable from a few rollouts of an untrained policy. Together they indicate where the method can help ($K > 1$) and whether the successor-combination rule matters at all (fan-out > 1).
5. An empirical study within the A-E PN ecosystem (Section 6): a comparison against discount-matched PPO and seven alternative credit rules on six problem configurations, an ablation ladder over the estimator family, and a negative control on which the mechanism predicts and obtains a null.

The findings are conditioned as follows. Causal-order estimation improves on discount-matched PPO at the four configurations with the largest K , by +0.19 to +0.55 in normalized return over 20 paired seeds, and is indistinguishable from it at the two with the smallest. Against other bounded credit rules that scope credit by component it is statistically indistinguishable, and we do not claim superiority over them. The predecessor-normalized variant loses its learned policy late in training, and the variant computing the causal estimand exactly is not the best member of the family. On the one benchmark with headroom below the anchor, those two variants and a third never learn at all (Section 6.7). Every statement about the expected policy gradient is conditional on Assumption 1, which is plausible for systems that decompose into weakly coupled sub-streams and not for systems in which one congested resource pool links all work (Section 5.3). All experiments are run within the A-E PN ecosystem, so replication elsewhere remains to be done.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 introduces A-E PN, the induced decision process and the provenance record. Section 4 defines the estimator and Section 5 analyses it. Section 6 reports the experiments, Section 7 discusses their interpretation, scope and limitations, and Section 8 concludes.

2. Related work

Deep RL for dynamic task assignment. A growing literature applies deep RL to scheduling, routing and business-process decision-making, typically encoding the instance as a graph consumed by a graph neural network: greedy graph-embedding heuristics (Dai et al., 2017), attention-based routing (Nazari et al., 2018) and graph-network dispatching for job-shop scheduling (Zhang et al., 2020); Mazyavkina et al. (2021) and Bengio et al. (2021) survey the

area. Within the A-E PN program, Lo Bianco et al. (2023) introduce the formalism, Lo Bianco et al. (2025b) a graph representation for infinite state and action spaces, and Lo Bianco et al. (2025a) a library supporting partial observability and multiple decisions. In all of this work the effort goes into the encoding and the policy architecture, and the simulator’s execution trace is not used as a learning signal. Our contribution is a credit-assignment mechanism, so it is compatible with and inherits from that literature.

Credit assignment in RL. Return decomposition (Arjona-Medina et al., 2019) redistributes return along a trajectory with a learned return model, and Hindsight Credit Assignment (Harutyunyan et al., 2019) reweights past actions by their learned likelihood of having caused an outcome. Both are learned redistributions. Token provenance is recorded exactly by the simulator, so the step that decides which rewards a decision may be credited with introduces no estimation error, although what is exact is the ancestry relation and not a causal effect (Section 3.2). Model-free counterfactual credit assignment (Mesnard et al., 2021) shares the motivation and builds low-variance baselines by conditioning on future events; simulator-replay counterfactual credit (CFPK) is the exact, model-based end of the same spectrum, at a simulation cost that CGAE-CF avoids.

Factored and multi-agent credit. Difference rewards (Wolpert and Tumer, 2001), COMA (Foerster et al., 2018) and value decomposition (Sunehag et al., 2018; Rashid et al., 2018) credit each agent against a counterfactual baseline or factor a joint value, and all require the entity partition to be supplied. MACCA (Ruan et al., 2024) learns the causal structure from offline data instead. The distinction we draw is therefore *learned versus recorded*: a structure estimated from transition tuples carries estimation error, whereas an A-E PN writes down, for each firing, which tokens were consumed and produced. Our partition also changes per episode with the realized case mix, and it factors a single-agent problem with many concurrent cases in the same way it would a multi-agent one.

The simulator as more than a source of samples. Che et al. (2024) make the simulator differentiable and take pathwise policy gradients through it; that needs smooth surrogate dynamics, where ours needs only recorded token identity, and the two are complementary. Yeganeh et al. (2026) control a semiconductor fab and start from an observation we share: because an action initiates an extended event that overlaps others, the next decision point may be uninformative for credit assignment. Their response groups events that co-occur in a time window and matches the group to one system-level reward; ours groups the decisions from whose outputs a reward’s tokens descend and needs only the rewards the episode already pays.

Decomposition in operations research. Dantzig–Wolfe (Dantzig and Wolfe, 1960) and Benders (Benders, 1962) decomposition split a structured problem along a block structure identified by hand. CGAE-CF applies the same idea to the policy-gradient return, along a block structure derived per episode from the simulator’s record.

Generalized advantage estimation and deliberate bias. Our estimator is GAE (Schulman et al., 2016) with one substitution, so GAE is the closest technical antecedent. GAE with $\lambda < 1$ is itself biased, and the biased settings

are the ones used in practice; its bias is a shrinkage by temporal distance, λ^k , where ours shrinks by causal distance. We therefore relocate a shrinkage that is already present rather than introduce one. Deliberate bias is common in this pipeline (PPO’s clipped surrogate, truncated importance weights, reward clipping), and in our experiments the unbiased member of the estimator family does not perform better than the biased one we ship.

Reward shaping and graph architectures. Potential-based shaping (Ng et al., 1999) cannot change the optimal policy; we use its SMDP-discounted form as one of three alternative routes tested in Appendix G. The policy is a Heterogeneous Graph Transformer (Hu et al., 2020), the standard choice for encoding combinatorial structures of variable size. That appendix also documents an architectural finding of independent interest: a per-node actor decoder over directed-only message-passing edges can be exactly blind to state outside its forward-reachable neighbourhood, a limitation the pooling critic does not share.

3. Background

This section describes the A-E PN formalism informally, defines the learning problem, and fixes the terminology used in the rest of the paper.

3.1. Action-Evolution Petri Nets and the induced decision process

A Petri net is a bipartite directed graph of *places* and *transitions*. Places hold *tokens*; in a colored net each token carries data, such as a task’s type or a resource’s identity. A transition is *enabled* when its input places hold a matching combination of tokens, called a *binding*, and *firing* it consumes that binding and produces tokens in its output places. The distribution of tokens over places, the *marking*, is the system state.

An A-E PN (Lo Bianco et al., 2023) partitions the transitions into two kinds and adds a global clock. *Evolution* transitions model the exogenous dynamics, such as arrivals and service completions; they fire automatically according to stochastic timed rules and advance the clock. *Action* transitions model the controllable decisions: whenever several action bindings are enabled at once, a *policy* selects which one to fire. Any firing may emit a scalar reward. Figure 1 shows a minimal assignment net.

Execution alternates between the environment evolving the marking and the agent choosing among enabled action bindings, which is a semi-Markov decision process (SMDP). We write s_d for the marking at decision d , a_d for the chosen binding and u_d for its clock. Delayed rewards are discounted by the SMDP factor $\rho_d(t) = e^{-\beta \max(0, t - u_d)}$. The policy π_θ is a graph neural network over the assignment-graph representation of Lo Bianco et al. (2025b), trained by PPO (Schulman et al., 2017) with GAE (Schulman et al., 2016). GAE forms $A_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}$ with $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, computed by the backward recursion $A_t = \delta_t + \gamma \lambda A_{t+1}$. Everything below changes only the successor $t + 1$ in this recursion.

Executing an A-E PN produces, for each firing, a record of the input tokens consumed, the output tokens produced, the firing time and the reward emitted. Writing $\mathcal{A}$ for the set of agent decisions in an episode, indexed in firing order,

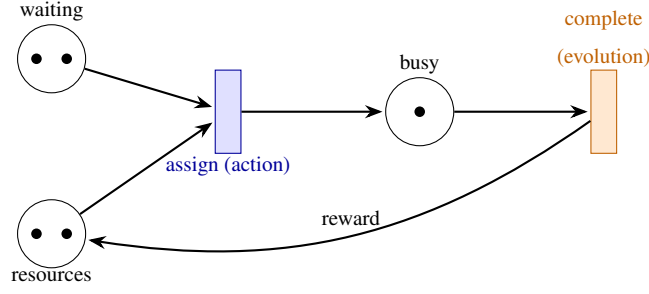


Figure 1: A minimal A-E PN for task assignment. The *assign* action transition (blue) is a decision point; the *complete* evolution transition (orange) fires after a stochastic delay, returns the resource and emits a reward.

the record induces a directed acyclic graph on $\mathcal{A}$: token identity links each consumed token back to the firing that produced it, transitively through evolution and forced firings. We call this the provenance DAG.

3.2. Provenance, causal ancestry and causal effect

Three notions must be kept apart. *Token provenance* is the recorded relation: composing the firing records by token identity gives, for any token, the firings from whose outputs it descends. It is exact and observational, a property of the executed trajectory. *Causal ancestry* is the relation we derive from it: decision d is a causal ancestor of reward r_j if some token consumed at j 's firing descends from an output token of d . Because a reward in an A-E PN is a deterministic function of the tokens consumed at its firing, ancestry is a necessary condition for d to have influenced r_j , and not a sufficient one: a decision may be an ancestor of a reward that would have been earned in the same amount under any other action. *Marginal causal effect* is what a counterfactual quantity such as a difference reward measures, and estimating it requires re-running the simulator or learning a model of the counterfactual. We do neither.

The estimator of Section 4 uses ancestry as a scoping device: it decides which rewards enter a decision's advantage and leaves the magnitude of credit to the ordinary policy-gradient machinery. Throughout, the word “causal” in *causal successor*, *causal component* and *causal depth* refers to this ancestry relation, never to an estimated effect size. The one place where causality in the stronger sense enters is Assumption 1, which is an assumption about the system and not something the record can establish.

4. Method: causal-order advantage estimation

This section defines the estimator in three steps: the successor relation, the ownership of rewards, and the recursion. Figure 2 illustrates the construction on a short episode.

Causal successors.. For a decision $d \in \mathcal{A}$, $s \in \text{succ}(d)$ when s consumes a token descending from an output token of d with no intervening decision. This is the *nearest producer* relation: evolution and forced firings are transparent, and the backward walk from a consumed token stops at the most recent decision on its provenance. We call $\text{succ}(d)$ the causal successors of d . The relation is read off the execution record and requires no token-value semantics.

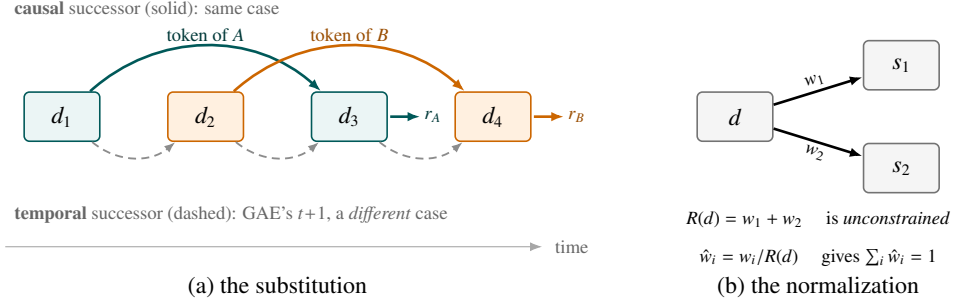


Figure 2: The mechanism. **(a)** Four decisions in wall-clock order, drawn from two interleaved cases A and B . Generalized advantage estimation bootstraps along the dashed *temporal* chain, so the TD error it charges to d_1 is the one produced at d_2 —a decision about a different case, whose reward d_1 did not influence. The token d_1 produced is instead consumed at d_3 , where the reward r_A is emitted. Causal-order estimation follows the solid provenance edges, so d_1 is credited with r_A and never with r_B . Reward ownership is a partition—each reward is assigned to the latest decision on its own lineage—so no reward is counted twice. **(b)** Where a decision’s tokens feed several later decisions, the edge weights must be normalized *over successors*. Token conservation constrains the sum over each node’s predecessors, which leaves the row sum $R(d)$ free; because $R(d)$ multiplies the bootstrapped value (Proposition 2), leaving it unbounded makes the recursion diverge in training.

Under token-flow postponement, a `POSTPONE` decision re-emits the whole marking and therefore stops the nearest-producer walk although it did not originate the flow. Making postponements transparent recovers provenance edges (measured at 97 per episode on the single-component environment) and yields a more faithful graph, but degrades learning (Appendix F). We report this as a measured boundary of the specification and keep postponements opaque.

Single-owner reward attribution. Let $\mathcal{A}(j)$ be the set of decisions in reward j ’s ancestry. Each reward r_j is assigned to the latest decision on its lineage, $\text{own}(j) = \arg \max_{d \in \mathcal{A}(j)} u_d$. Ownership is a partition: every reward is counted exactly once. This is what makes the estimator a return decomposition rather than a hindsight Q -sample, in which every decision on a lineage would receive the reward’s full mass.

The recursion. Let $w(d \rightarrow s)$ be the share of s ’s consumed tokens that descend from d , so that token conservation gives

$$\sum_{d \in \text{pred}(s)} w(d \rightarrow s) = 1. \quad (1)$$

Define the row sum and the successor-normalized weights

$$R(d) = \sum_{s \in \text{succ}(d)} w(d \rightarrow s), \quad \hat{w}(d \rightarrow s) = \frac{w(d \rightarrow s)}{R(d)}, \quad \sum_{s \in \text{succ}(d)} \hat{w}(d \rightarrow s) = 1, \quad (2)$$

with $\hat{w}(d \rightarrow s) = 1/|\text{succ}(d)|$ when $R(d) = 0$. With $\rho(\Delta t) = e^{-\beta \Delta t}$ the SMDP discount applied at causal depth, the estimator is

$$A[d] = \text{owned}[d] - V[d] + \sum_{s \in \text{succ}(d)} \hat{w}(d \rightarrow s) \rho(u_s - u_d) (V[s] + \lambda A[s]), \quad (3)$$

Table 1: The estimator family, and the two sums that matter.

variant	$c(d \rightarrow s)$	$\sum_{\text{pred}(s)} c$	$\sum_{\text{succ}(d)} c$
CGAE-DAG	1 (closure, set semantics)	—	convex
CGAE-CF	w/R	unbounded	= 1 exactly
CGAE	$1/ \text{succ}(d) $	unbounded	= 1 exactly
CGAE-CAP	$w/\max(R, 1)$	≤ 1	≤ 1
CGAE-F	w	= 1 exactly	unbounded

evaluated by one backward pass over $\mathcal{A}$ in decreasing clock order and emitted as $Q[d] = A[d] + V[d]$. It is consumed like every other credit scheme: the policy uses $A_d = Q[d] - V(s_d)$ and the value head regresses on $Q[d]$, so the critic converges to the object the recursion bootstraps through. Algorithm 1 in Appendix B states the procedure line by line.

Three properties follow directly. Where every causal edge has out-degree one, $\hat{w} \equiv 1$ and (3) reduces to ordinary SMDP-GAE, which we verify to floating point on real traces. Under equal inflow shares $\hat{w} = 1/|\text{succ}(d)|$, so the mean-over-successors rule (CGAE) is the equal-share instance of (3). A POSTPONE decision receives the SMDP temporal-difference advantage $e^{-\beta\tau} V(s') - V(s)$, so that every arm supports postponement in the same way. The computation costs about one millisecond per episode, under 8% of a training iteration.

5. Theoretical analysis

This section places CGAE-CF in a family of estimators, identifies the member that is unbiased for the policy gradient and the assumptions this needs, shows why the bootstrap coefficient must be bounded, quantifies the variance reduction, and bounds the bias of the estimator we ship relative to the unbiased member. Proofs are in Appendix A.

5.1. A common form for the estimator family

Every variant we consider is a choice of edge coefficient $c(d \rightarrow s)$ in (3). Unrolling with $\lambda = 1$ and $V \equiv 0$,

$$Q[d] = \sum_{j: \text{own}(j) \in \text{desc}^*(d)} c_d(j) \rho(u_j - u_d) r_j, \quad c_d(j) = \sum_{\text{paths } d \rightsquigarrow \text{own}(j)} \prod c, \quad (4)$$

so a variant is characterized by the total path weight it places on each descendant’s owned reward. Table 1 lists the family.

5.2. The causal partition, the estimand and its unbiasedness

Write $\text{desc}(d)$ for the set of decisions reachable from d under the successor relation, $\text{desc}^*(d) = \text{desc}(d) \cup \{d\}$, and $\text{anc}^*(e) = \{d : e \in \text{desc}^*(d)\}$. Separately, build an undirected graph on $\mathcal{A}$ with an edge (d, d') whenever some reward has both on its provenance; its connected components are the *causal components*, and $c(d)$ is the component of d . Every reward with non-empty ancestry has a well-defined component $c(j)$, and rewards with empty ancestry are constants that drop out of the gradient. Note that $\text{desc}^*(d) \subseteq c(d)$.

Definition 1 (The structural statistic K). K is the number of causal components containing at least one reward, computed per episode and averaged over rollouts of an untrained policy. It is a real number rather than an integer, which is why the values 2.0, 3.3 and 6.3 reported for the N -copies benchmark sit at or below N . It requires no training and no reward model.

For decision d let

$$G_d^{\text{full}} = \sum_{j: t_j \geq u_d} \rho_d(t_j) r_j, \quad \Delta_d = \sum_{\substack{j: t_j \geq u_d \\ c(j) \neq c(d)}} \rho_d(t_j) r_j,$$

so that G_d^{full} is the ordinary SMDP return-to-go used by PPO and by the lineage ablation mc-Q, and Δ_d is the part of it generated by other components. Let $\mathcal{F}_d$ be the history up to but excluding a_d . We take the policy-gradient theorem in its causality form as given: the estimator $\sum_d \nabla \log \pi(a_d | s_d)(G_d^{\text{full}} - V(s_d))$ is unbiased for $\nabla_{\theta} J$.

Assumption 1 (Component mean-exogeneity). For every decision d , $\mathbb{E}[\Delta_d | \mathcal{F}_d, a_d] = \mathbb{E}[\Delta_d | \mathcal{F}_d]$: given the pre-action history, the expected cross-component future reward does not depend on the action taken at d .

We refer to this assumption as (A1). A-E PN semantics supply part of the argument for it. A reward is a deterministic function of the tokens consumed at its firing, so along the realized execution r_j can have been influenced by a_d only if $d \in \mathcal{A}(j)$, hence only if $c(j) = c(d)$. But (A1) concerns the conditional distribution over executions, including those that a different action would have induced. Two channels can break it. The exogenous randomness driving cross-component events must be, in conditional mean, unaffected by a_d ; this holds under independent per-firing draws, and Remark 3 treats a single shared random stream. The substantive channel is *contention*: if the action at d occupies or releases a resource that work in another component would otherwise have used, that component’s reward stream shifts in distribution even though no token links the two. The record cannot detect this, because the coupling acts through what did not happen. Whether (A1) is tenable is therefore a question about the system, and Section 5.3 discusses it.

The closure estimand.. The first row of Table 1 is the $c \equiv 1$ member: the $\lambda = 1$, $V \equiv 0$ estimator crediting d with each descendant’s owned reward exactly once,

$$A^*[d] = \sum_{j: \text{own}(j) \in \text{desc}^*(d)} \rho_d(t_j) r_j. \quad (5)$$

CGAE-DAG computes it by construction, which we verify against the definition to 0 units in the last place over 191 decisions. Because ownership is a partition, each reward enters (5) at most once even when its owner is reachable from d along several paths.

Unbiasedness needs two further conditions, which are verifiable on the trace.

Assumption 2 (Time-forward causal edges). $u_s > u_d$ for every causal edge $d \rightarrow s$.

Assumption 3 (Single-origin rewards). For every reward r_j with $\mathcal{A}(j) \neq \emptyset$, the tokens consumed at j ’s firing have a single nearest-producer decision a_j .

Assumption 2 makes the provenance DAG acyclic and makes the discount telescope along any path. Assumption 3 holds whenever a reward is earned by completing one unit of work whose most recent handling was a single decision, which is the ordinary case in task assignment. It fails for a *join* reward, earned by a firing that consumes tokens last touched by two different decisions. On all six configurations of this paper it holds on every reward (Section 6).

Lemma 1 (Ownership dominates the ancestry). *Under Assumptions 2 and 3, $\text{own}(j) = a_j$ and $\mathcal{A}(j) \subseteq \text{anc}^*(\text{own}(j))$. Consequently, for every decision d , $\text{own}(j) \notin \text{desc}^*(d)$ implies $\mathcal{A}(j) \cap \text{desc}^*(d) = \emptyset$.*

Proposition 1 (The closure estimand is unbiased). *Under Assumptions 1, 2 and 3, the estimator $\hat{g}^* = \sum_d \nabla_\theta \log \pi_\theta(a_d | s_d)(A^*[d] - V(s_d))$ satisfies $\mathbb{E}[\hat{g}^*] = \nabla_\theta J(\theta)$ for any $\mathcal{F}_d$ -measurable baseline V .*

Proposition 1 licenses (5) as a reference point and every relative statement made below. It says nothing about performance: CGAE-DAG computes (5) exactly and is one of the worst arms we measured. The proof applies mean-exogeneity in a slightly broader form than (A1) states, extending it to rewards in d 's own component whose owner lies outside the descendant closure; Lemma 1 places those rewards outside d 's provenance descendants entirely, so the only channel by which a_d could move them is the contention channel that threatens (A1) itself. Where Assumption 3 fails, the broader form is genuinely stronger.

5.3. The domain of validity of Assumption 1

Call a system *decomposable* at the scale of an episode when the work in flight partitions into sub-streams drawing on disjoint capacity, so that an assignment inside one sub-stream neither consumes nor delays capacity available to another. Call it *strongly coupled* when a single contended resource serves all work, so that every assignment changes when that resource next becomes free to everything else. (A1) is a natural modelling assumption at the first extreme and is not credible at the second. Real systems lie between, and their position is a design variable: the resource-pooling and flexibility literature studies exactly this dial, since flexible capacity buys performance by creating the cross-stream coupling that (A1) rules out (Jordan and Graves, 1995).

The condition is approximately met wherever capacity is segmented and the segments are not close to saturation: multi-site service operations with site-dedicated staff, parallel production lines with their own operators, regional fleets, wards with ring-fenced beds, picking zones with zone-assigned pickers. It fails wherever sub-streams share capacity, and more generally under congestion: a shared queue, a finite buffer that can block an upstream stage, rework that returns work to a common queue, cross-trained staff who float between sites. Even in the favourable cases, escalation paths and overflow rules reintroduce coupling, and (A1) is stated per decision, so it can hold for most decisions in an episode and fail for the few that touch a shared resource. Our single-component environment (Appendix D) is a deliberate instance of the coupled extreme.

K (Definition 1) measures how far an episode's realized provenance decomposes. $K > 1$ is necessary for (A1) to have content, but not sufficient, because K is computed from executions that did happen and cannot see the contention channel. We therefore treat K as a screening statistic and the assessment of (A1) as a modelling judgement informed

by it. Which results depend on (A1) is stated precisely: Propositions 1 and 3 hold only under it, and so does every statement phrased relative to the closure estimand. Propositions 2 and 4, Lemma 2 and Proposition 5 are algebraic identities about the recursion and hold regardless. If (A1) fails, CGAE-CF remains a bounded estimator whose departure from a well-defined reference is controlled, but its expected gradient is no longer that of the objective. The one datum we have on this case, the $K = 1$ negative control, shows a null rather than a degradation relative to PPO.

5.4. Why the bootstrap coefficient must be bounded

Proposition 2 (Successor normalization). *For the recursion (3) with coefficient c , the coefficient multiplying the bootstrap term at d is $\sum_{s \in \text{succ}(d)} c(d \rightarrow s)$. The predecessor constraint (1) does not bound this quantity. Writing $\hat{G}(d) = \sum_s \hat{w} \rho(V[s] + \lambda A[s])$ for the estimator’s own bootstrapped continuation value, the unnormalized and normalized forms differ by*

$$A_{\text{CGAE-F}}[d] - A_{\text{CGAE-CF}}[d] = (R(d) - 1) \hat{G}(d). \quad (6)$$

The proof is immediate from $w = R\hat{w}$. On the single-component environment $R(d)$ has mean 1.008 and standard deviation 0.542 over the range $[0.143, 3.429]$ (Figure C.6 in Appendix C), so a mean of about 1 conceals a dispersion problem. Three consequences follow. First, the error scales with the critic, not the reward: $\hat{G}(d) = O(V) = O(H\bar{r})$ for remaining causal horizon H and mean per-decision reward $\bar{r}$, while the action-attributable signal is $\text{owned}[d] = O(\bar{r})$, so the estimator degrades as the critic improves. Second, the error does not cancel as a baseline: $R(d)$ depends on the realized DAG downstream of d , hence on the action at d ; forking every action at a decision state under common random numbers, R is identical across actions at only 18.8% (four copies) and 6.2% (single component) of fork points. Third, successor normalization removes the error identically, since $\sum_s \hat{w} = 1$ gives $R \equiv 1$.

Remark 1 (The predecessor sum is a choice of estimand, not a defect). *Reading (1) as a second conservation requirement would admit only $c = w / \max(R, 1)$; the policy-gradient argument does not require it, since nothing there constrains advantages across different decisions to sum to anything. $\sum_{\text{pred}(s)} c > 1$ means each of s ’s joint ancestors is credited with the whole continuation, the but-for reading behind difference-reward methods. Empirically the distinction barely moves total credit mass ($\sum_d Q[d]/G$ is 4.74 for CGAE-F against 4.86 for CGAE-CF on the single-component environment), and forcing the predecessor sum down (CGAE-CAP) costs performance.*

5.5. Variance reduction and the role of K

The estimand (5) discards Δ_d without moving the expected gradient. What it buys is variance, and the size of the purchase is governed by K . The following isolates that mechanism in the simplest model exhibiting it.

Proposition 3 (Component-scoped variance). *Fix a decision d and suppose the future reward-to-go splits as $G_d^{\text{full}} = \sum_{k=1}^K R_k$ with d in component 1, where R_k is component k ’s discounted reward-to-go and, conditionally on s_d , the R_k are mutually independent with variances σ_k^2 . Let the baseline be any s_d -measurable $V(s_d)$. Then*

$$\text{Var}(G_d^{\text{full}} - V(s_d) \mid s_d) = \sum_{k=1}^K \sigma_k^2, \quad \text{Var}(R_1 - V(s_d) \mid s_d) = \sigma_1^2.$$

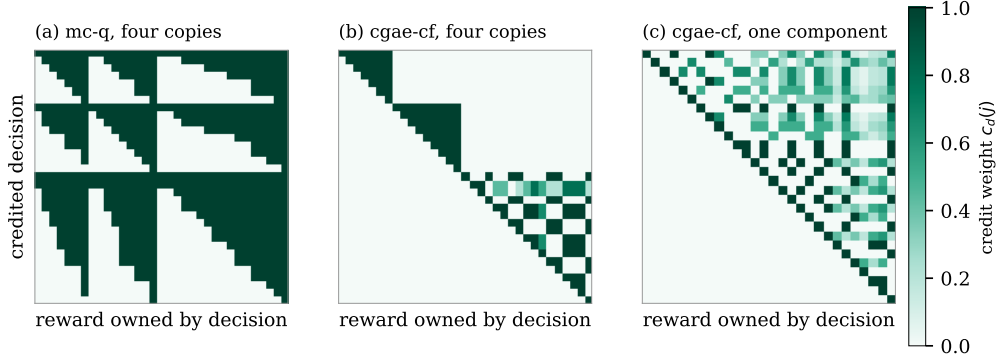


Figure 3: The credit matrix $c_d(j)$, the weight decision d places on the reward owned by decision j , on single episodes. (a) and (b) are the same episode of the four-copy environment under the same ordering: return-to-go credits a decision for essentially everything after it, while causal-order credit is confined to its descendants and is block structured, the blocks being the three reward-bearing components the estimator found without being given a partition. (c) is the negative control: one block, so scoping removes nothing independent.

If the components have comparable variance, component-scoped credit reduces the per-decision advantage variance by a factor of about K .

The scoping is observable. With $S(d) = |\text{desc}(d)|/|\{\text{decisions after } d\}|$, balanced components give $S \approx 1/K$; on the scaling benchmark S measures 1.000, 0.472, 0.221 and 0.114 for $N = 1, 2, 4, 8$ against $1/N$. At $N = 1$ the estimator provably performs no scoping and every method ties. S is not K , however: the single-component environment has $S = 0.474$, equal to the two-copy benchmark’s 0.472, yet gains nothing, because scoping away half the timeline reduces variance only if what is removed is independent of what is kept.

Crediting every causal ancestor inflates credit relative to the return $G = \sum_j r_j$, but far less than plain return-to-go does, which enters a reward at decision k into all k preceding advantages. Measured, $\sum_d Q[d]/G$ is 39.78, 21.23 and 18.40 for return-to-go on the three main environments against 4.34, 4.32 and 4.86 for cgae-cf, a four- to ninefold reduction tracking $1/S$. Figure 3 shows that the causal estimator is not only sparser but block sparse, the blocks being the components it discovered.

5.6. A bias bound for cgae-cf

Proposition 2 says the unnormalized form is unusable; it does not say the normalized form is correct. This subsection bounds how far cgae-cf departs from the closure estimand A^* . The bound is relative: it says how much bias is added on top of an estimator that is unbiased under (A1).

Assumption 4 (Bounded score). $\|\nabla \log \pi(a | s)\| \leq B$ for all (s, a) visited.

Lemma 2 (Path weights are visit probabilities). *Under Assumption 2, $\hat{W} = [\hat{w}(d \rightarrow s)]$ is row-stochastic on non-sink nodes and induces a Markov chain on $\mathcal{A}$ with sinks absorbing. For this chain, $c_d(e)$ is the probability that the walk started at d visits e , so $c_d(e) \in [0, 1]$.*

Define the *re-convergence deficiency* at d in worst-case and reward-weighted form,

$$\delta_d = \max_{e \in \text{desc}(d)} (1 - c_d(e)), \quad \bar{\delta}_d = \frac{\sum_{e \in \text{desc}^*(d)} (1 - c_d(e)) \rho_{de} |\text{owned}[e]|}{\sum_{e \in \text{desc}^*(d)} \rho_{de} |\text{owned}[e]|},$$

and $\delta = \max_d \delta_d$. Both are computable from a few rollouts without training.

Proposition 4 (Bounded bias). *Under Assumptions 2 and 4, with $\lambda = 1$ and $V \equiv 0$:*

- (i) $c_d(e) \in [0, 1]$ for every d and every $e \in \text{desc}(d)$;
- (ii) $|A[d] - A^*[d]| \leq \bar{\delta}_d \sum_{e \in \text{desc}^*(d)} |\text{owned}[e]| \leq \delta_d \sum_{e \in \text{desc}^*(d)} |\text{owned}[e]|$;
- (iii) the policy-gradient bias relative to the A^* estimator obeys $\|\nabla J_{\text{CGAE-CF}} - \nabla J_\star\| \leq B \delta \sum_e |\text{owned}[e]| \cdot |\text{anc}^*(e)| \leq B \delta n G_{\text{abs}}$, with $G_{\text{abs}} = \sum_j |r_j|$;
- (iv) if every node of $\text{desc}^*(d)$ has out-degree at most one, then $\delta_d = 0$ and $A[d] = A^*[d]$ exactly; more generally $\delta_d = 0$ if and only if $\text{desc}^*(d)$ is totally ordered by reachability.

The bound is instrumented. Table C.6 in Appendix C reports $\bar{\delta}$ per environment over 100 untrained rollouts. On multi-site routing every decision has out-degree one, so $\delta = 0$ and CGAE-CF coincides with the closure estimand. Where out-degree exceeds one the deficiency grows with fan-out: the mean of $1 - c_d(e)$ over strict descendant pairs is 0.165 at four copies, 0.188 at eight and 0.569 on the single-component environment, and the reward-weighted $\bar{\delta}$ is smaller throughout (0.079, 0.073, 0.408). Lemma 2 is verified directly: $\hat{W}$ is row-stochastic to 2.2×10^{-16} , $c_d(e)$ never leaves $[0, 1]$ over 21,073 causal edges, and $\sum_e c_d(e) \text{owned}[e]$ reproduces the recursion’s output to 1.8×10^{-15} . The proposition bounds bias, not performance: CGAE-DAG attains $\delta = 0$ by construction and performs poorly. It explains why CGAE-CF is safe, not why it is good.

Two remarks complete the analysis. With a critic and $\lambda < 1$ the recursion is exactly GAE in expectation over the causal walk, so the additional bias is the ordinary bootstrap bias of an imperfect critic. And when owned rewards are nonnegative, as in every environment here, CGAE-CF’s departure from A^* is a per-decision rescaling $A[d] = m_d A^*[d]$ with $m_d \in [0, 1]$. A uniform rescaling cannot change the ascent direction and is removed by the per-batch advantage standardization, so only the dispersion of m_d can distort the gradient, through an exact cosine law in its coefficient of variation. Appendix C states, proves and measures this: the credit direction is preserved to within 8.8% in the worst case and exactly on multi-site.

6. Experiments

This section compares CGAE-CF against discount-matched PPO and seven alternative credit rules on six problem configurations. The protocol and the environments are described first, then the results.

6.1. Protocol

All comparisons use common random numbers: every greedy evaluation point, for every seed and method, scores on one fixed scenario set. Returns are normalized as $(\text{score} - \text{random})/(\text{heuristic} - \text{random})$, where the anchors are

Table 2: Structural statistics, computed from random rollouts before training. K = reward-bearing causal components; fan-out = $\text{mean } |\text{succ}(d)|$; S = scoping ratio.

environment	K	top comp. share	fan-out	S	character
multi-site routing	8.0	15.5%	1.000	0.106	realistic, most decomposable
$N = 8$ copies	6.3	35.5%	1.160	0.114	scaling endpoint
$N = 4$ copies	3.3	42.4%	1.137	0.221	scaling benchmark
$N = 4$ copies, hard	3.1	68.3%	1.295	0.179	scaling benchmark, unsaturated
$N = 2$ copies	2.0	56.1%	1.020	0.472	low- K point
single component	1.0	100%	1.387	0.474	negative control

a random policy and a simple priority rule. We report paired t -tests and Wilcoxon signed-rank tests over 20 seeds, and count *collapses* (normalized final ≤ 0.25). Every arm uses identical hyperparameters with no per-arm tuning; Appendix E gives them together with the architecture and the evaluation protocol, and Appendix B defines every comparator arm.

All results are reported at a 40-epoch budget. An earlier protocol trained the scaling benchmark for 15 epochs, at which the policy has barely moved (final entropy 0.91–0.94 from 1.00), and it produced a significant but spurious result: at two copies CGAE-CF measured 0.401 against CGAE-F’s 0.837 ($p = 0.0043$), which at 40 epochs becomes +0.102 ($p = 0.51$), sign reversed. Appendix F documents this convergence-speed effect; the 15-epoch cells are retained only as a budget-sensitivity ablation.

6.2. Environments and their structure

We use three environment families, all specified in Appendix D. The first is a multi-site skills-based routing problem with dedicated per-site specialists, which is the realistic case. The second is an N -copies scaling benchmark of N structurally independent assignment problems sharing one policy, in which the independence structure is a controlled parameter; it is run in a base form, whose anchor is provably optimal, and in a harder form with headroom above the anchor. The third is a two-stage service line whose single shared employee pool links every case into one causal component, which is the negative control. Table 2 gives their structural statistics.

Two features of Table 2 are used repeatedly. K predicts whether scoping can help, and fan-out predicts whether the successor-combination rule matters at all. On multi-site the fan-out is exactly 1.000, so CGAE-CF, CGAE, CGAE-CAP and CGAE-F are bit-identical there, with identical learning curves on every seed, which we verify rather than assume.

The structural conditions of Section 5.2 are checked on 20 untrained rollouts of each environment. No causal edge points backwards in time (0 violations over 21,073 edges). Every reward firing consumes its tokens from a single nearest-producer decision (3,157 of 3,157 rewards on the four original environments, 915 of 915 on the hard variant), and that decision is $\text{own}(j)$ in every case, as Lemma 1 requires. The lemma’s consequence was tested directly as

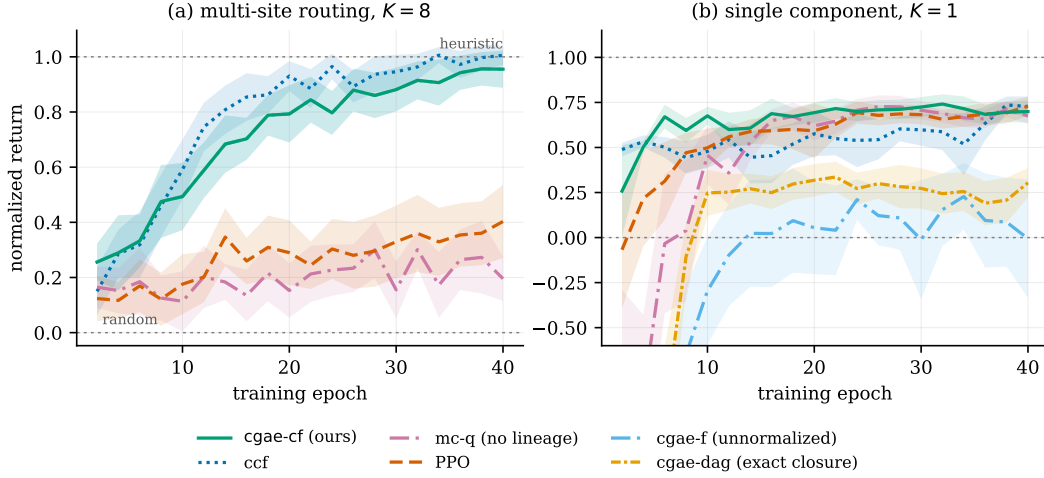


Figure 4: Learning curves, mean over 20 common-random-number seeds with 95% confidence bands; both panels show the same six arms and share the legend. **(a)** Where the problem decomposes ($K=8$), causal-order credit separates from PPO by the tenth epoch and reaches the heuristic, while PPO plateaus below 0.4 and the lineage ablation mc-q stays at or below PPO. On this environment the fan-out is 1, so CGAE-F is bit-identical to CGAE-CF and its curve lies exactly under it; CGAE-DAG was not run here. **(b)** Where the problem does not decompose ($K=1$), CGAE-CF, CCF, MC-Q and PPO are indistinguishable; CGAE-DAG plateaus at a third of the attainable return, and CGAE-F climbs, turns over and ends near random, the late deterioration Proposition 2 anticipates.

Table 3: Normalized final return at 40 epochs, 20 common-random-number seeds per cell, as mean (standard deviation) [collapsed seeds out of 20]. Dashes are cells not run; bold marks an arm that is significantly best in its row. On multi-site the causal fan-out is exactly 1, so CGAE-CF, CGAE, CGAE-CAP and CGAE-F are the same algorithm and their columns are identical.

environment	K	CGAE-CF	CGAE	CCF	CGAE-CAP	CGAE-F	MC-Q	PPO
multi-site	8.0	0.955 (.15)[0]	0.955 (.15)[0]	1.005 (.08)[0]	0.955 (.15)[0]	0.955 (.15)[0]	0.196 (.18)[11]	0.403 (.30)[6]
$N=8$	6.3	0.999 (.03)[0]	—	0.977 (.06)[0]	—	0.741 (.65)[3]	—	0.804 (.23)[0]
$N=4$	3.3	0.980 (.09)[0]	0.995 (.03)[0]	0.974 (.10)[0]	0.971 (.10)[0]	0.989 (.03)[0]	0.562 (.29)[5]	0.715 (.27)[11]
$N=4$ hard	3.1	0.405 (.09)[1]	0.388 (.13)[4]	0.351 (.10)[2]	0.422 (.12)[2]	-1.588 (.48)[19]	0.099 (.15)[18]	0.183 (.13)[15]
$N=2$	2.0	0.882 (.18)[1]	0.797 (.30)[3]	0.799 (.30)[3]	0.761 (.43)[3]	0.751 (.34)[4]	0.651 (.39)[5]	0.802 (.30)[3]
single comp.	1.0	0.699 (.14)[0]	0.643 (.19)[1]	0.727 (.13)[0]	0.722 (.15)[0]	-0.006 (.75)[10]	0.675 (.10)[0]	0.729 (.12)[0]

well: across 177,537 (decision, reward) pairs whose owner lies outside the descendant closure, and 48,210 on the hard variant, not one has an ancestry decision inside it. The check is not vacuous, since rewards have 4.35 to 10.85 ancestry decisions on average, so the configuration Assumption 3 excludes is available in these nets and does not arise.

6.3. Main results

Figure 4 shows the learning curves at the two ends of the K range and Table 3 the final numbers; Table C.7 in the appendix gives the paired tests in full. Four observations summarize them.

Against PPO the effect is large where K is large.. On multi-site cgae-cf improves normalized return by +0.552 (19W/1L, $p < 0.0001$) and collapses on 0 of 20 seeds against PPO’s 6. The paired gains are +0.194 ($p = 0.0019$) at eight copies, +0.264 ($p = 0.0009$) at four and +0.222 ($p < 0.0001$) on the hard four-copy variant, where PPO collapses on 15 of 20 seeds. At two copies (+0.080, $p = 0.34$) and at $K=1$ (−0.030) the effect is absent. The effect is therefore present at the four high- K configurations and absent at the two low- K ones, as Proposition 3 predicts. The multi-site collapse is not a tuning artifact: a one-knob grid around the shared setting (learning rate, GAE λ , entropy bonus; 20 seeds each, at eight episodes per epoch) leaves PPO between 0.15 and 0.28 with 9 to 14 seeds collapsed in every cell, none better than the shared setting (Appendix E).

The gain is located in the causal restriction.. The lineage ablation mc-q is the same SMDP return-to-go with the causal restriction removed and everything else held fixed. It loses to cgae-cf wherever scoping is possible, by margins tracking K : +0.760 on multi-site (20W/0L), +0.418 at four copies and +0.306 on the hard variant (all $p < 0.0001$). At two copies the difference is +0.230 with $p = 0.010$ on the paired t -test but 0.08 after Holm correction and 0.11 on the signed-rank test, so we count it as suggestive only. Where there is nothing to scope it is null (+0.024, $p = 0.52$). On multi-site the ablation even finishes below PPO (−0.207, $p = 0.018$).

Against other bounded causal schemes the effect is null.. Versus cgae , ccf , cgae-cap and cgae-f the paired difference spans −0.015 to +0.009 at four copies, −0.040 to +0.056 on the single-component environment and +0.080 to +0.204 at two copies, with every $p \geq 0.27$ except the $N=2$ minimum of 0.052. On the hard variant the spread against cgae , ccf and cgae-cap is −0.017 to +0.054 with every $p \geq 0.11$, and on multi-site ccf is numerically ahead (−0.050, $p = 0.21$). An earlier version of this work, at a shorter budget and fewer seeds, reported a significant advantage over ccf on multi-site; at the stated protocol the sign reverses and the difference is null.

On the negative control the method is null, as predicted.. On the single-component environment cgae-cf is −0.030 against PPO ($p = 0.46$) and seven arms lie within 0.06 of one another: $K = 1.0$, one component holds all the reward mass, and there is nothing to scope. This environment is also where the ablations separate, as the next subsection shows.

6.4. The successor normalization is necessary

Table 4 reports the ablation ladder on the negative control. Two rungs are significant: the predecessor-normalized form cgae-f and the exact-closure form cgae-dag perform substantially worse, while the choice between flow-weighted, uniform and sub-stochastic handling does not separate the remaining arms.

cgae-f deteriorates with training, as Proposition 2 anticipates. Averaged over the same 20 seeds it reaches a best-checkpoint value of 0.443 and ends at −0.006, a within-run drop of 0.449 against 0.089–0.136 for every arm that learns. Ten of its seeds end collapsed while only two are collapsed at their best checkpoint, and one seed’s greedy curve rises, holds for six evaluation points, then falls while policy entropy stays flat at 0.18: a low-entropy poor policy

Table 4: Single-component environment ($K=1$), 20 seeds, normalized final return, paired against CGAE-CF. Only two rungs of the ladder are significant.

variant	final (SD)	collapses	paired vs CGAE-CF
CFGAE	0.739 _(0.11)	0	−0.040 ($p = 0.27$)
PPO	0.729 _(0.12)	0	−0.030 ($p = 0.46$)
CCF	0.727 _(0.13)	0	−0.028 ($p = 0.54$)
CGAE-CAP	0.722 _(0.15)	0	−0.023 ($p = 0.55$)
CGAE-CF	0.699 _(0.14)	0	—
MC-Q	0.675 _(0.10)	0	+0.024 ($p = 0.52$)
CGAE	0.643 _(0.19)	1	+0.056 ($p = 0.36$)
CGAE-DAG	0.304 _(0.19)	8	+0.395 ($p < 0.0001$)
CGAE-F	−0.006 _(0.75)	10	+0.705 ($p = 0.0006$)

rather than a failure to explore. This is the direction an error term proportional to the critic would produce, and we report it as consistent with the mechanism rather than as an isolation of it. The same arm is also unsafe at the highest- K configuration: at eight copies it measures 0.741 ± 0.654 with 3 collapsed seeds against CGAE-CF’s 0.999 ± 0.025 and none, and the failure is invisible at five seeds because it appears as occasional failed runs (Figure F.7 in Appendix F). The ordering is unchanged under a best-checkpoint reading.

6.5. Exactness of the estimand and empirical performance

CGAE-DAG computes (5) by construction and is the unbiased member of the family under (A1). On the single-component environment it ranks second-worst of the nine arms, and on the hard four-copy variant it ranks last (Section 6.7). One structural reason is visible in Table 2: on a single-component environment the descendant closure is most of the remaining episode ($S = 0.474$), so the exact estimand approaches the unscoped return and exactness buys no scoping.

The ordering of the four working arms is not accounted for by any mechanism we tested. Six candidates were probed and all failed, two of them inverted: per-state ranking fidelity against a common-random-number counterfactual puts CGAE-DAG first, and credit signal-to-noise ratio likewise favours the lower-performing arm. Appendix F reports each probe with the measurement that rejected it. The only variable that separates the learning from the deteriorating arms without overlap is final policy entropy (0.27–0.56 against 0.13–0.15), whose causal status is untested. What the ladder does support is a narrower claim: within this family, exactness of the estimand is not what distinguishes the arms that learn from the one that deteriorates, whereas boundedness of the bootstrap coefficient coincides with that division on the single-component environment. Section 6.7 shows that boundedness does not predict a second kind of failure.

6.6. How far is the anchor from optimal?

Normalized return is measured against a myopic heuristic, so a score of 1 means *as good as that rule*, not *optimal*. On two of the three families the gap between them can be computed exactly, and we do so.

A single copy of the N -copies net and a single site of the multi-site net are small enough to solve by backward induction: the state is the clock, the queue counts by type and the remaining service time of each server; service times are bounded integers and arrivals are one per unit time, so the recursion is exact. Because the reported configurations share nothing across copies or sites, the N -copy and four-site optima are N and 4 times the single-unit optimum. Three clock conventions are not fixed by the environment source; we enumerate all eight combinations, evaluate the heuristic under each, and keep the convention reproducing the simulator’s own heuristic, which matches it to +0.028 tasks per copy and +0.059 per site. Two optima are relevant. The anchors never postpone, so the *non-idling* optimum says how good the normalization’s unit is. On N -copies and the single-component environment the learned arms may postpone (Table E.9), so their ceiling is the unrestricted optimum; on multi-site postponement is disabled for learners and anchors alike, so there the non-idling optimum is the ceiling.

On N -copies the heuristic is exactly optimal: matched-first non-idling attains 11.1245 expected completions per copy and so does the non-idling optimum, as the exchange argument for shortest expected processing time on a single server predicts. Allowing idling adds 0.0326 per copy, all of it a horizon artifact recovered by idling from $t \geq 17$ only. On the normalized scale the anchor sits at 1.000 and the ceiling for the arms is 1.010. On multi-site the single-site non-idling optimum is 20.2892 against the heuristic’s 20.1643, so the ceiling is 1.029; idling would raise it to 1.104, but no arm may idle there.

Table 3 can therefore be read as an optimality gap. Against the ceiling of 1.010, CGAE-CF attains 98.9% of the achievable return at $N=8$, 97.0% at $N=4$ and 87.3% at $N=2$, against PPO’s 79.6%, 70.8% and 79.4%; at $N=8$ it closes essentially the whole remaining gap to a computed optimum. Two consequences run the other way. First, a benchmark whose optimum is attained by most arms cannot discriminate among them: at $N=8$ and $N=4$ the causal arms occupy 0.971–0.999, so the nulls between CGAE-CF and the other bounded schemes there are evidence of saturation, not of equivalence. Second, multi-site is closer to the same condition than its larger headroom suggests: CGAE-CF ’s 0.955 captures 92.8% of the achievable improvement and CCF ’s 1.005 captures 97.7%, against PPO’s 39.2%, so the causal arms reach the three-line rule that anchors the scale and the null between them is again partly a ceiling effect. Of the six configurations, only the hard four-copy variant leaves every arm well below its ceiling, which is why we ran it.

6.7. An unsaturated scaling benchmark

The hard variant makes the per-copy problem a genuine assignment rather than a binary match: each copy has three task types and two heterogeneous specialists, one fast on each of the first two types (delay 1), while the third type matches neither specialist (delay 3) and a cross-assignment is slower still (delay 4), each plus $U\{0, 1\}$ (Appendix D). The copies remain independent by construction, with $K = 3.1$, fan-out 1.295 and scoping ratio 0.179 (Table 2). The matched-first anchor is not optimal here, so the ceiling exceeds 1 by a margin we have not computed; what

Table 5: Hard four-copy variant ($K=3.1$), 20 common-random-number seeds, normalized final return as mean (SD) [collapsed seeds], the paired difference of CGAE-CF minus each arm, and each arm’s paired difference against PPO with wins out of 20.

arm	final (SD)	collapses	CGAE-CF minus arm	arm minus PPO	
CGAE-CAP	0.422 (0.12)	2	-0.017 ($p = 0.60$)	$+0.239$ ($p < 0.0001$)	18W
CGAE-CF	0.405 (0.09)	1	—	$+0.222$ ($p < 0.0001$)	17W
CGAE	0.388 (0.13)	4	$+0.017$ ($p = 0.50$)	$+0.205$ ($p = 0.0002$)	16W
CCF	0.351 (0.10)	2	$+0.054$ ($p = 0.11$)	$+0.168$ ($p < 0.0001$)	18W
PPO	0.183 (0.13)	15	$+0.222$ ($p < 0.0001$)	—	
MC-Q	0.099 (0.15)	18	$+0.306$ ($p < 0.0001$)	-0.084 ($p = 0.07$)	9W
CFGAE	-1.361 (0.49)	20	$+1.766$ ($p < 0.0001$)	-1.544 ($p < 0.0001$)	1W
CGAE-DAG	-1.486 (0.28)	20	$+1.891$ ($p < 0.0001$)	-1.669 ($p < 0.0001$)	0W
CGAE-F	-1.588 (0.48)	19	$+1.993$ ($p < 0.0001$)	-1.771 ($p < 0.0001$)	1W

matters is that no arm approaches it, the best final value being 0.422. The protocol is that of the other N -copies cells (Appendix E).

Table 5 reports all nine arms. With headroom, the gap to PPO is large and the intra-family nulls persist. PPO collapses on 15 of 20 seeds and ends at 0.183, and the four bounded causal schemes beat it by $+0.168$ to $+0.239$ on 16 to 18 of 20 seeds ($p \leq 0.0002$). The lineage ablation is again null against PPO and loses to CGAE-CF on all 20 seeds. Among the four schemes that learn the spread is 0.071 and no pairwise difference is significant (CGAE-CF against CCF $+0.054$, $p = 0.11$; against CGAE and CGAE-CAP within ± 0.017). With more than half the scale unclaimed, the bounded variants remain indistinguishable, so their null on the base family is not attributable to saturation alone.

Three arms never learn, and boundedness does not predict which. CGAE-F, CGAE-DAG and CFGAE end below random on 19, 20 and 19 seeds. This is not the late deterioration of Section 6.4: at the first evaluation, after three epochs, CGAE-DAG already returns exactly zero on all 20 seeds, CGAE-F on 11 and CFGAE on 7, which means the argmax policy postpones at essentially every decision. A static diagnostic points at why. Under a random policy and a constant critic, the standardized advantage each scheme assigns to postpone decisions on this variant is -1.05 to -1.27 for CGAE-CF, CGAE-CAP and CGAE, with no postpone decision receiving a positive advantage, but $+0.46$ for CGAE-F and $+0.32$ for CGAE-DAG; on the base variant every scheme gives postpone a negative advantage. The two cgae-family arms that fail are thus the two whose credit tells the untrained policy that idling pays. We read this as consistent with the failure rather than as its cause, since CCF also gives postpone a mildly positive advantage ($+0.21$) yet learns, and CFGAE is not covered by the diagnostic. Two of the three failing arms have a bounded bootstrap coefficient, so the observation of Section 6.4 holds on the single-component environment and does not extend to this failure mode.

At 40 epochs every arm that learns is still rising slowly (final policy entropy 0.88–0.90), so these are fixed-budget numbers rather than converged ones. The gap between CGAE-CF and PPO lies between 0.20 and 0.25 at each of the last six evaluation points, while the ordering within the family may still move.

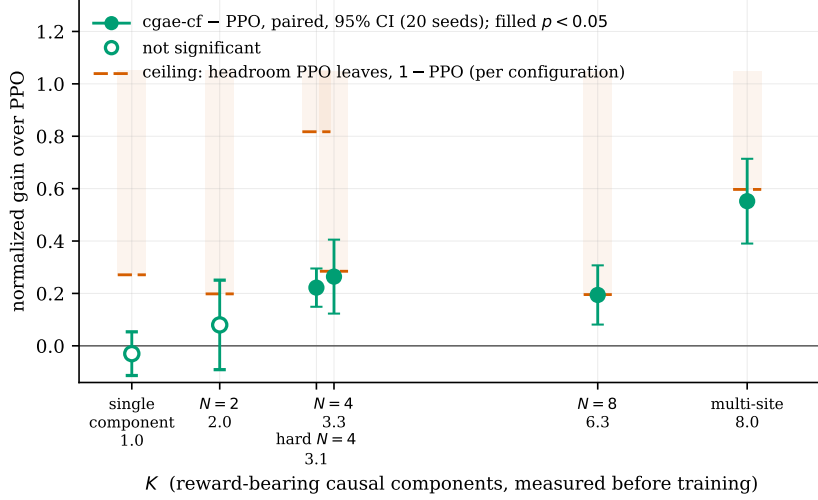


Figure 5: Paired gain over PPO against K , with the headroom PPO itself leaves drawn per configuration as a ceiling. The gain is absent at $K=1$, significant at the four largest K , and positive but not significant at $K=2.0$. At two and eight copies PPO already reaches 0.80, and the measured gain sits essentially on the bound (+0.194 against a ceiling of 0.196 at eight copies); at multi-site PPO leaves 0.597 unclaimed, of which the method claims 0.552. We fit no trend through the six points.

6.8. Scaling

The measured gain over PPO is positive at every $K > 1$, significant at the four largest values of K , and absent at $K = 1$. It is not monotone in K , and Figure 5 shows why: much of the departure is due to headroom. At $N=8$ every causal arm saturates near the heuristic and PPO reaches 0.805, so the headroom above PPO to the ceiling of 1.010 is 0.206 and *cgae-cf* claims 0.194 of it; at $N=2$ the ceiling is similarly low and the gain does not reach significance. The hard four-copy variant shows the converse: at nearly the same K as the base one, PPO leaves 0.817 unclaimed and the gain is +0.222, so headroom bounds the gain without predicting it. The defensible claim is therefore that K predicts *whether* component-scoped credit can help, not *how much*. The $N=2$ configuration serves as the low- K end of this curve and for nothing else: at the converged budget the spread across all seven arms is 0.169 with nothing significant.

7. Discussion

7.1. How the empirical gains should be interpreted

Three things are measured rather than interpreted. Causal-order credit improves normalized return over discount-matched PPO on the four configurations with the largest K and is indistinguishable from it on the two with the smallest. The lineage ablation, which removes only the provenance restriction, loses to *cgae-cf* wherever scoping is possible and is null where it is not, which places the gain in the restriction rather than in the discounting or the estimator form. And the predecessor-normalized variant deteriorates late in training on two environments, while every bounded

variant does not; on the hard variant, however, two bounded arms fail from the first evaluation, so boundedness rules out the late deterioration and not every failure.

That the gain is caused by variance reduction of the kind Proposition 3 describes is a reading of this evidence rather than a measurement of it. The pattern is consistent with the proposition, but the proposition assumes conditional independence across components, and the effect sizes are not a clean function of K . At least two channels plausibly contribute and our design does not separate them. The first is *decomposition*: restricting a decision’s advantage to rewards it could have influenced removes terms that, under (A1), have zero expected contribution and nonzero variance. The second is *boundedness*: successor normalization keeps the credit vector a bounded, low-dispersion object regardless of the critic, and scoping cuts total credit mass four- to ninefold, which changes the conditioning of the value-regression target. The ablation ladder speaks mainly to the second channel. The most defensible statement is therefore that restricting credit along recorded provenance produces the gain over PPO, which the ablation establishes, and that among schemes applying such a restriction, keeping the estimator bounded matters more than making it exact. Why the bounded variants order as they do is not settled.

7.2. A decision rule for practitioners

Compute K from a handful of random rollouts before adopting anything. If the problem decomposes, even partially, into weakly coupled sub-streams, such as parallel lines, multiple sites or segmented resource pools, causal-order credit offers a cheap drop-in gain with no hand-specified structure. If $K \approx 1$, expect a null; on our negative control the method is neither better nor worse than the baseline, though one environment is thin evidence for safety in general. Compute mean fan-out as well: where it is 1, every variant in Table 1 is the same algorithm. Read K alongside a judgement about resource sharing (Section 5.3), since K sees only the provenance of executions that happened.

7.3. Generality beyond Action-Evolution Petri Nets

The method is not a modification of RL that applies to an arbitrary environment. It uses four properties of the formalism: work is carried by individually identified tokens; every state change is a firing that records which tokens it consumed and produced, so provenance composes through merges and splits; rewards are emitted by firings that consume tokens, so each reward has a well-defined ancestry; and decisions are themselves firings in the same record. Colored Petri nets (Jensen and Kristensen, 2009) satisfy the first three by definition and need only a marking of which firings are decisions. Entity-flow discrete-event simulators identify entities individually, so an entity-level record is obtainable in principle, but the standard trace records an entity’s path through blocks rather than the input–output relation of each event, so batch, assemble and separate operations are where a token-level record would have to be added. Process-mining event logs carry a case identifier, which is coarser than token provenance but enough to recover the same partition where rewards are earned at case completion and cases do not interact. Models whose state is aggregate counts do not expose the relation at all. The scope boundary is the reward structure rather than the domain: the estimator needs rewards attached to token-consuming firings and enough reward mass in distinct components for

K to exceed one. Admission control and revenue management satisfy this directly; per-period holding costs would have to be discretized onto firings first. We have not run a second domain.

7.4. Limitations

Provenance and decomposability. The method cannot be applied where the simulator does not record token identity and firing-level input–output relations. Instrumenting an existing simulator is usually straightforward but not free, and its cost outside A-E PN implementations is untested. The gain requires reward mass in more than one causal component, failing which the estimator reduces to an unscoped return-to-go. Its value is therefore a property of the instance rather than of the method.

Near-optimal anchors and near-saturated benchmarks. On N -copies the anchor is the non-idling optimum and most arms sit within 0.04 of the ceiling, so the nulls between CGAE-CF and the other bounded rules there say the benchmark has no resolving power left. The hard variant removes the ceiling and still does not separate them at 20 seeds, which is evidence of a small or absent difference, but its budget is not converged and its optimum is uncomputed. On multi-site the causal arms reach 0.955–1.005 against a ceiling of 1.029, so its clear result is the gap to PPO rather than any ordering among the causal schemes, and matching a three-line priority rule is not evidence of operational competitiveness beyond it. The optimum of the single-component environment is not computed.

Join rewards. Proposition 1 needs Assumption 3, and a reward earned by a firing that consumes tokens last handled by two different decisions violates it. The estimator still runs, so the failure is silent, which is why we report the condition as checkable and check it. None of our environments is a merge or assembly problem; a manufacturing net paying a reward on joining two sub-assemblies is where it would fail. The condition concerns the reward structure, so it is independent of (A1) and of K .

Assumption 1 is not verified. (A1) is argued from A-E PN semantics plus a modelling judgement about contention, and we know of no cheap test for it. Where it fails, Propositions 1 and 3 lapse; the algebraic results are unaffected but are then relative to an estimand whose own status has weakened. Our evidence about the coupled regime rests on a single environment at $K = 1$, on which the method is null. We have not measured intermediate coupling, where (A1) is violated but $K > 1$, which is arguably the most common situation in practice and the one where a biased scoping rule could do harm. The flexible-worker knob of Appendix D is designed to sweep this dial and is not exercised here.

Empirical scope. All experiments are conducted within the A-E PN ecosystem: six configurations from three families, one library, one graph representation, one architecture and one hyperparameter setting. The configurations share a short horizon, unit completion rewards and a throughput objective, and two of the three families are synthetic. We can conclude that provenance-scoped credit helps in decomposable A-E PN task-assignment problems under this protocol and that the predecessor-normalized form is unsafe in it; we cannot conclude that the ordering transfers to

other encodings, longer horizons, cost objectives or other discrete-event formalisms. Among the bounded variants we cannot say which to prefer or why.

7.5. *Three alternative routes on the negative control*

Proposition 3 predicts a null at $K = 1$, which raises the question whether the residual gap on a single-component problem can be closed by giving the learner more structure by some other route. We tested three, none of which touches credit assignment: potential-based reward shaping read off the net’s topology, structural input features derived from the conflict graph, and a fix to the actor architecture. All three are neutral or harmful on the bottleneck environment. Shaping is policy-invariant (Ng et al., 1999) yet harmful at large coefficients, because constant exogenous arrivals make the potential a noisy value-regression target at a finite budget. Features that are a function of action type alone are provably redundant, since every action node already carries its type. The architectural fix repairs a real blindness of the per-node actor decoder but is neutral here, because the shared pool leaves the unfixed architecture a marginal path to the same information. Appendix G reports all three. The difficulty of this environment lies neither in what the learner can observe nor in an avoidable credit bias, and we state it as an open problem.

7.6. *Future work*

The natural next steps are a coupling sweep that varies shared capacity continuously and measures performance against (A1) violation directly; replication in a non-Petri-net discrete-event simulator instrumented to record entity-level provenance; a cost-minimization problem with mixed-sign, per-period rewards; and an interventional test of the ablation ordering, an entropy-bonus manipulation being the most direct.

8. Conclusion

This paper showed that the token provenance an A-E PN simulation records as a by-product of execution can replace a hand-specified factorization of credit with a derived one. Running generalized advantage estimation along the provenance DAG rather than the trajectory index yields a cheap, fork-free policy gradient that improved on discount-matched PPO by +0.19 to +0.55 in normalized return on the configurations whose reward generation is at least partially decomposable, and was neither better nor worse where it is not. On the scaling benchmark, where the optimum is computable, the method reaches 98.9% of it where PPO reaches 79.6%. The properties that make this possible, identified tokens, firing-level input–output records, rewards emitted by token-consuming firings and decisions that are themselves firings, are shared by other colored-net and entity-flow simulators, though we have not tested it outside this one. Two qualifications travel with the result: the statements about the expected policy gradient hold under a component mean-exogeneity assumption that segmented capacity makes plausible and shared capacity does not; and within the estimator family the design choice that mattered empirically was keeping the bootstrap coefficient bounded rather than computing the causal estimand exactly, a distinction we can document but have not explained. A third concerns the benchmarks: on the base scaling family and on the realistic configuration the causal arms reach a ceiling within a

few percent of the heuristic that anchors the scale, so those benchmarks separate causal credit from its absence but not the causal schemes from one another, and only the harder scaling variant leaves room, where they still do not separate at twenty seeds.

Appendix A. Proofs

The statements are in Section 5; only the arguments are collected here.

Lemma 1. Let a_j be the single nearest producer supplied by Assumption 3. Every token consumed at j 's firing descends from an output token of a_j with no intervening decision, so the backward walk reaches any other member of $\mathcal{A}(j)$ only by continuing past a_j , through a_j 's own consumed tokens. The nearest-producer relation composed transitively is the successor relation of Section 4, so those decisions are precisely the successor-ancestors of a_j : $\mathcal{A}(j) \subseteq \text{anc}^*(a_j)$. By Assumption 2 every successor edge strictly increases the clock, hence $u_{a_j} \geq u_{d'}$ for every $d' \in \mathcal{A}(j)$, with equality only at $d' = a_j$; since $\text{own}(j)$ is by definition the clock-latest member of $\mathcal{A}(j)$, $\text{own}(j) = a_j$.

For the consequence, suppose some $d' \in \mathcal{A}(j)$ lay in $\text{desc}^*(d)$. From $\mathcal{A}(j) \subseteq \text{anc}^*(\text{own}(j))$ we have $d' \in \text{anc}^*(\text{own}(j))$, i.e. $\text{own}(j) \in \text{desc}^*(d')$, and desc^* is transitive, so $\text{own}(j) \in \text{desc}^*(d)$ — contradicting the hypothesis. Hence $\mathcal{A}(j) \cap \text{desc}^*(d) = \emptyset$. $\square$

Proposition 1. Write the discarded mass as $G_d^{\text{full}} - A^*[d] = \Delta_d + \Xi_d$, where Δ_d collects the cross-component rewards and $\Xi_d = \sum \rho_d(t_j)r_j$ runs over the rewards j with $t_j \geq u_d$, $c(j) = c(d)$ and $\text{own}(j) \notin \text{desc}^*(d)$: those in d 's own component that the closure nonetheless omits. It suffices to show each vanishes in the gradient.

Fix d and condition on $\mathcal{F}_d$. Since a_d is drawn given $\mathcal{F}_d$ and both quantities are realized afterwards,

$$\mathbb{E}[\nabla \log \pi_\theta(a_d | s_d) X_d | \mathcal{F}_d] = \mathbb{E}_{a_d}[\nabla \log \pi_\theta(a_d | s_d) \mathbb{E}[X_d | \mathcal{F}_d, a_d]]$$

for $X_d \in \{\Delta_d, \Xi_d\}$. For $X_d = \Delta_d$ the inner expectation is free of a_d by Assumption 1. For $X_d = \Xi_d$, a reward j in that sum has $\text{own}(j) \notin \text{desc}^*(d)$, so by Lemma 1 $\mathcal{A}(j) \cap \text{desc}^*(d) = \emptyset$: no decision of j 's ancestry is a descendant of d , hence no descendant of a_d lies on j 's provenance. Along the realized execution r_j is therefore not influenced by a_d , since r_j is a deterministic function of the tokens consumed at its firing and none of them descends from a_d . Passing from that statement to $\mathbb{E}[\Xi_d | \mathcal{F}_d, a_d] = \mathbb{E}[\Xi_d | \mathcal{F}_d]$ requires the same mean-exogeneity condition as for Δ_d , now applied to within-component rewards whose owner falls outside the closure; we assume it in that form, and Section 5.3 discusses the channels that would break it, which are the same in both cases.

The step through Lemma 1 is where Assumption 3 is consumed, and it cannot be replaced by a time-order argument. Ownership is the clock-latest decision of $\mathcal{A}(j)$, which on its own does *not* prevent an *earlier* member of $\mathcal{A}(j)$ from lying in $\text{desc}^*(d)$ while the owner does not: that is exactly the configuration a join reward creates, and there r_j is a genuine provenance descendant of a_d sitting in Ξ_d , where the argument above would assume it away. Assumption 3 excludes it.

In both cases the inner expectation is some $\bar{X}_d$ not depending on a_d , so the outer expectation is $\bar{X}_d \mathbb{E}_{a_d}[\nabla \log \pi_\theta(a_d | s_d) | \mathcal{F}_d] = \bar{X}_d \nabla_\theta \sum_a \pi_\theta(a | s_d) = \bar{X}_d \nabla_\theta 1 = 0$ by the score-function identity. The baseline term contributes zero by the same identity. Summing over d and taking total expectation gives the claim. $\square$

Proposition 3. Conditionally on s_d the baseline is a constant, so it does not affect either variance. By conditional independence $\text{Var}(\sum_k R_k \mid s_d) = \sum_k \text{Var}(R_k \mid s_d) = \sum_k \sigma_k^2$, which is the first equality; the second is immediate. Under $\sigma_k^2 \approx \sigma^2$ the ratio is $K\sigma^2/\sigma^2 = K$, and standard deviation scales as the square root. $\square$

Lemma 2. Row-stochasticity is (2). By Assumption 2 the DAG is acyclic, so a walk visits any node at most once; consequently the events {the walk’s initial segment is π }, indexed by the distinct paths $\pi : d \rightsquigarrow e$, are pairwise disjoint and their union is exactly the event that the walk visits e . Each has probability $\prod_{(x \rightarrow y) \in \pi} \hat{w}(x \rightarrow y)$, and summing gives $c_d(e)$. A probability lies in $[0, 1]$. $\square$

Proposition 4. (i) is Lemma 2. For (ii), telescoping of ρ gives $A^*[d] - A[d] = \sum_e (1 - c_d(e)) \rho_{de} \text{owned}[e]$; take absolute values, use $\rho \leq 1$ and $0 \leq 1 - c_d(e) \leq \delta_d$, and note the middle expression is exactly $\bar{\delta}_d$ times the total owned mass. For (iii), the two estimators differ only through the summand, so $\nabla J_{\text{CGAE-CF}} - \nabla J_\star = \mathbb{E}[\sum_d (A[d] - A^*[d]) \nabla \log \pi_d]$; apply the triangle inequality, then (ii) and Assumption 4, and reindex the double sum by owner, each e being counted once per ancestor. For (iv), if out-degree is at most one throughout $\text{desc}^*(d)$ the walk is deterministic and visits every descendant, so $c_d(e) = 1$; conversely $c_d(e) = 1$ for all e says every maximal walk from d meets every descendant, which is total order by reachability. $\square$

Appendix B. Estimator definitions and additional remarks

Every arm compared in Section 6 differs from every other only in how the per-decision target $Q[d]$ is formed. All of them are consumed identically—the policy uses $A_d = Q[d] - V(s_d)$ and the value head regresses on $Q[d]$ —so the comparison isolates the credit rule and nothing else. Throughout, u_d is decision d ’s clock, t_j reward j ’s firing time, $\rho_d(t) = e^{-\beta \max(0, t - u_d)}$ the SMDP discount, $\mathcal{A}(j)$ the decisions on j ’s lineage, $c(\cdot)$ the causal-component map and $\text{own}(\cdot)$, $\text{succ}(\cdot)$, $\text{desc}^*(\cdot)$ as in Sections 4 and 5.2.

Non-causal reference points..

- **PPO** is the unmodified baseline: ordinary GAE along the trajectory index with the SMDP discount, $A_t = \delta_t + \gamma \lambda A_{t+1}$.
- **mc-Q** (*the lineage ablation*) is the plain Monte-Carlo SMDP return-to-go, $Q[d] = \sum_{j: t_j \geq u_d} \rho_d(t_j) r_j$. Every decision sees every subsequent reward whether or not it caused it. It shares the estimator *form* of the causal Monte-Carlo schemes ($\lambda = 1$, no bootstrap, wall-clock discount) and differs from them only by dropping the causal restriction, which is exactly what makes it the ablation that prices the lineage. It needs no token DAG.

Component-scoped schemes.. These restrict *which* rewards a decision sees, keeping the Monte-Carlo return-to-go.

- **LRQ** credits d with the full discounted mass of every reward on whose lineage it sits, $Q[d] = \sum_{j: d \in \mathcal{A}(j)} \rho_d(t_j) r_j$. A reward with k lineage decisions contributes its whole mass to all k of them, so this is a per-decision hindsight

Q -sample and *not* a return decomposition; it must never be summed as a return. It is the scheme single-owner attribution (Section 4) replaces.

- ccf builds the realized causal-component partition of Section 5.2 by union-find over each reward’s lineage, then credits $Q[d] = \sum_{j: c(j)=c(d), t_j \geq u_d} \rho_d(t_j) r_j$: its own component’s return-to-go. Because the partition is computed from the *realized* episode, component membership can itself depend on the action taken.
- s-ccf replaces that realized partition with a static, topology-only one derived from the net’s structure, which is action-invariant by construction. It is reported here for completeness; it does not appear in the main tables.

Component-filtered GAE. cFGAE takes ccf’s partition but consumes it through PPO’s own recursion rather than as a Monte-Carlo sample: for each component it runs the ordinary SMDP-GAE over the step-reward stream masked to that component, and each decision reads the advantage from its own component’s pass. Its $K = 1$ limit is therefore PPO *exactly*—the mask is identically one and the recursion is term-for-term PPO’s—whereas every Monte-Carlo scheme above degenerates at $K = 1$ to mc-q, a different estimator that merely happens to score similarly. That exactness is why cFGAE is the right comparator for "does the *partition* help, holding the estimator form fixed".

The causal-order family. These run GAE’s recursion along the provenance DAG, differing only in the edge coefficient $c(d \rightarrow s)$ of (3); Table 1 collects the coefficients and the two sums that matter.

- CGAE-CF (*the method*) uses $\hat{w} = w/R$, normalized over successors.
- CGAE uses $1/|\text{succ}(d)|$, the equal-share special case.
- CGAE-F uses the raw flow weight w , normalized over predecessors—the natural formulation, and the one Proposition 2 shows to be unbounded.
- CGAE-CAP uses $w/\max(R, 1)$, which only ever divides, so both the predecessor and successor sums are at most one. It prices the share-versus-but-for choice of Remark 1.
- CGAE-DAG abandons path weighting and uses set semantics: it credits each decision with each descendant’s owned reward exactly once, computing A^* of (5) directly. It is the unbiased member of the family (Proposition 1).
- CGAE-CF2 is not an ablation of any design choice but a different DAG construction: CGAE-CF with POSTPONE made transparent to the nearest-producer walk. It appears only in the discussion of that specification wrinkle.

Postponement. In every causal scheme a POSTPONE decision receives, in place of a lineage credit, the SMDP temporal-difference advantage $e^{-\beta\tau} V(s') - V(s)$, so that every arm gives postponement consistent support.

Remark 2 (Why single-owner attribution rather than LRQ’s lineage sample). LRQ gives each reward’s full mass to every decision on its lineage, so summing its output over decisions overcounts the return by a factor that grows with

Algorithm 1 Causal-order advantage estimation (CGAE-CF)

```
1: input: decisions  $\mathcal{A}$  with clocks  $u_d$ ; token history; critic values  $V[\cdot]$ ;  $\lambda, \beta$ 
2: build  $\text{succ}(\cdot)$  and  $w(\cdot \rightarrow \cdot)$ : one backward nearest-producer walk per consumed token
3:  $\text{owned}[d] \leftarrow 0$  for all  $d$ 
4: for all rewards  $r_j$  do
5:   if  $\mathcal{A}(j) = \emptyset$  then
6:     continue ▷ exogenous: enters no decision's credit
7:   end if
8:    $\text{owned}[\text{own}(j)] += r_j$  ▷ latest decision on  $j$ 's lineage
9: end for
10: for all  $d \in \mathcal{A}$  in decreasing  $u_d$  do
11:    $S \leftarrow \text{succ}(d) \setminus \{d\}$ 
12:   if  $S = \emptyset$  then
13:      $A[d] \leftarrow \text{owned}[d] - V[d]$  ▷ causal sink
14:   else
15:      $R \leftarrow \sum_{s \in S} w(d \rightarrow s)$ 
16:     if  $R > 0$  then
17:        $\hat{w}(d \rightarrow s) \leftarrow w(d \rightarrow s)/R$  for  $s \in S$ 
18:     else
19:        $\hat{w}(d \rightarrow s) \leftarrow 1/|S|$  ▷ no attributable inflow: the equal-share case
20:     end if
21:      $A[d] \leftarrow \text{owned}[d] - V[d] + \sum_{s \in S} \hat{w}(d \rightarrow s) \rho(u_s - u_d) (V[s] + \lambda A[s])$ 
22:   end if
23: end for
24: return  $Q[d] = A[d] + V[d]$  for all  $d$ 
```

lineage depth. Single-owner attribution makes ownership a partition, which is what turns the estimator into a return decomposition and lets the unrolled form (4) be read as a weighting of each reward. It also removes a bias LRQ carries in the opposite direction: restricting credit to one lineage drops the rewards that sibling lineages—the roads not taken at d —would have earned, and those are causal descendants of alternative actions at d , hence not mean-independent of a_d .

Remark 3 (Shared random streams and Assumption 1). With a single global random stream, changing a_d can shift the order in which cross-component events consume random draws, so their realized values may differ across counterfactual actions. This affects the realized Δ_d but not $\mathbb{E}[\Delta_d \mid \mathcal{F}_d, a_d]$, so Assumption 1—which is a statement about conditional means—is unaffected. The measurements in Section 6 that fork the simulator under common random

numbers exploit exactly this distinction.

Appendix C. The relative bound and the cosine law

Section 5.6 summarizes this appendix. Proposition 4(iii) bounds an absolute quantity whose bound scales with n , and no bound on the *ratio* of gradient norms can exist, since near a stationary point the denominator vanishes while the numerator need not. The useful relative statement is about *direction*, and for this estimator it is available in closed form.

Assumption 5 (Nonnegative owned rewards). $\text{owned}[e] \geq 0$ for all e .

This holds in every environment we study—throughput and completion rewards are nonnegative, and we verified that no decision receives negative owned reward in any measured episode—but it is a property of the reward model, not a theorem, and estimators for cost-minimization problems with mixed-sign rewards would need the general form of Proposition 4(ii) instead.

Recall the *reward-weighted deficiency* $\bar{\delta}_d$ of Section 5.6, whose absolute values Assumption 5 makes redundant, and define its multiplier

$$\bar{\delta}_d = \frac{\sum_e (1 - c_d(e)) \rho_{de} \text{owned}[e]}{\sum_e \rho_{de} \text{owned}[e]}, \quad m_d = 1 - \bar{\delta}_d \in [0, 1],$$

for every d with $A^\star[d] \neq 0$.

Proposition 5 (Exact rescaling and an exact cosine law). *Under Assumptions 2 and 5, with $\lambda = 1$, $V \equiv 0$:*

- (i) **Exact rescaling.** $A[d] = m_d A^\star[d]$ for every d .
- (ii) **The level of the shrinkage is free.** If $m_d \equiv m$ is constant, then $\nabla J_{\text{CGAE-CF}} = m \nabla J_\star$: the ascent direction and the set of stationary points are identical, and the factor m is absorbed by the step size—and removed outright by the per-batch advantage standardization used throughout. Only the dispersion of m_d can distort the direction.
- (iii) **Cosine law.** Let $p_d \propto A^\star[d]^2$ and let CV be the coefficient of variation of m under p . Then $\cos \angle(A, A^\star) = (1 + \text{CV}^2)^{-1/2}$ exactly, hence $1 - \cos \angle \leq \frac{1}{2} \text{CV}^2$.
- (iv) CV is dimensionless, independent of n , and computable from a handful of untrained rollouts.

Proposition 5. (i) By Assumption 5 the denominator of $\bar{\delta}_d$ is $A^\star[d]$ and its numerator is $A^\star[d] - A[d]$, so $\bar{\delta}_d = 1 - A[d]/A^\star[d]$, which rearranges to (i). (ii) is immediate from linearity of ∇J in the per-decision coefficient. For (iii), write $u = A^\star$ and $A_d = m_d u_d$. Then $\langle A, u \rangle = \sum_d m_d u_d^2 = \mathbb{E}_p[m] \|u\|^2$ and $\|A\|^2 = \sum_d m_d^2 u_d^2 = \mathbb{E}_p[m^2] \|u\|^2$, so

$$\cos \angle = \frac{\mathbb{E}_p[m]}{\sqrt{\mathbb{E}_p[m^2]}} = \frac{\mathbb{E}_p[m]}{\sqrt{\mathbb{E}_p[m]^2 + \text{Var}_p(m)}} = (1 + \text{CV}^2)^{-1/2},$$

and $(1 + x)^{-1/2} \geq 1 - x/2$ gives the bound. □

Table C.6: Proposition 5 measured on 100 untrained rollouts per environment ($\lambda = 1$, $V \equiv 0$, $\beta = 0$), pooled over all decisions with $A^* \neq 0$. CV is the A^* -weighted coefficient of variation of the multiplier m_d . Computed by `_diag_cosine_law.py`.

environment	$\bar{\delta}$ (mean)	CV	$\cos \angle$ measured	$(1 + \text{CV}^2)^{-1/2}$	$1 - \cos \angle$
multi-site	0.000	0.000	1.00000	1.00000	0.000
$N=4$	0.079	0.370	0.93784	0.93784	0.062
$N=8$	0.073	0.438	0.91611	0.91611	0.084
single comp.	0.408	0.451	0.91156	0.91156	0.088

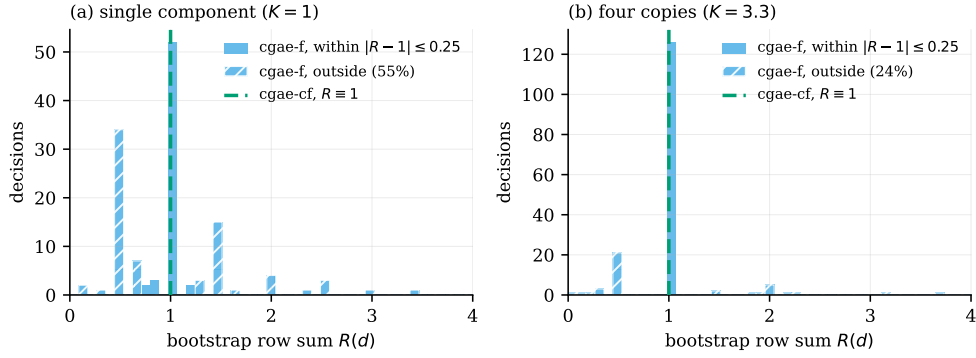


Figure C.6: The bootstrap row sum $R(d)$ over decisions of five random rollouts. CGAE-F leaves it unconstrained—mean 1.008, standard deviation 0.542, range [0.143, 3.429] on the single-component environment—so (6) does not vanish; the hatched mass is the 55.3% of decisions (24.1% at four copies) where it exceeds a quarter of the critic. CGAE-CF divides by R , so its coefficient is the dashed rule at exactly 1 for every decision. The aggregate mean of ≈ 1 is what makes the defect easy to overlook: it is a dispersion problem, not a bias in the mean.

Measured.. Table C.6 reports both sides. The identity (i) holds to machine precision (maximum absolute deviation 1.8×10^{-15} over all episodes), and the cosine law (iii) is exact to 2.2×10^{-16} —these are verifications of algebra, not fits, and the last two columns agreeing to five decimals in every row is the whole content of the check. The practical content is the last column: the direction of the credit vector is preserved to within 8.8% in the worst case, and *exactly* on multi-site, where out-degree is one so $\bar{\delta} \equiv 0$. The $\bar{\delta}$ column is the ρ -weighted deficiency of Section 5.6, with the absolute values dropped by Assumption 5; at $\beta = 0$, $\rho \equiv 1$.

Remark 4 (What the cosine law does and does not cover). *Proposition 5(iii) is a statement about the vectors of per-decision coefficients A and A^* , not directly about the gradients $\sum_d A_d \nabla \log \pi_d$. The two coincide when the score vectors $\{\nabla \log \pi_d\}$ are orthonormal; in general they differ by a factor governed by the conditioning of the score Gram matrix, which we do not bound. We therefore read Table C.6 as characterizing the estimator, not as a guarantee on the realized gradient. As with Proposition 4, everything is stated relative to A^* , whose unbiasedness rests on Assumption 1.*

Remark 5 (The gap between the model and the estimator). *Two idealizations separate Proposition 3 from what CGAE-*

Table C.7: Paired comparisons against `cgae-cf` (20 seeds, paired t -test). The pattern is uniform on the five base configurations: large and significant against non-causal or unbounded-coefficient baselines, null against every bounded causal variant. The hard four-copy variant keeps the first half of that pattern and breaks the second only for the two bounded arms that never learn there (Section 6.7).

environment	comparator	difference	p	
multi-site	PPO	+0.552	< 0.0001	***
multi-site	MC-Q	+0.760	< 0.0001	***
multi-site	CCF	−0.050	0.21	null
$N=8$	PPO	+0.194	0.0019	**
$N=8$	CGAE-F	+0.258	0.0999	
$N=8$	CCF	+0.021	0.24	null
$N=4$	MC-Q	+0.418	< 0.0001	***
$N=4$	CFGAE	+0.370	0.0005	***
$N=4$	PPO	+0.264	0.0009	***
$N=4$	CGAE, CCF, CGAE-CAP, CGAE-F	−0.015 to +0.009	≥ 0.43	null
$N=4$ hard	CGAE-F, CGAE-DAG, CFGAE	+1.77 to +1.99	< 0.0001	***
$N=4$ hard	MC-Q	+0.306	< 0.0001	***
$N=4$ hard	PPO	+0.222	< 0.0001	***
$N=4$ hard	CCF	+0.054	0.11	null
$N=4$ hard	CGAE, CGAE-CAP	−0.017 to +0.017	≥ 0.50	null
$N=2$	MC-Q	+0.230	0.010	not after Holm (0.08)
$N=2$	all other arms	+0.080 to +0.204	≥ 0.052	null
single comp.	CGAE-F	+0.705	0.0006	***
single comp.	CGAE-DAG	+0.395	< 0.0001	***
single comp.	PPO, CGAE, CGAE-CAP, CFGAE, CCF, MC-Q	−0.040 to +0.056	≥ 0.27	null

`cf` actually computes. Conditional independence across components is stronger than Assumption 1, which constrains only conditional means: (A1) is what buys unbiasedness, independence is what buys the variance factor, and an environment can satisfy the first while only approximately satisfying the second. And `cgae-cf` scopes to $\text{desc}^*(d)$, which is contained in $c(d)$ and usually strictly smaller, so it discards more than the model does—which is why the empirical gain is not a clean function of K (Section 6.8). The proposition should be read as identifying the mechanism and its governing quantity, not as predicting an effect size.

Appendix D. Environments

All three environment families are Action-Evolution Petri Nets built with the same library and solved by the same agent; they differ in topology, not in machinery. Each grants reward 1 per completed task, so return is throughput over the horizon. Table D.8 collects the parameters.

Multi-site skills-based routing (the realistic environment). A firm runs four service sites. Each site has its own stochastic arrival stream of tasks of two types and its own heterogeneous pool of dedicated specialists: one skilled for each type, fast on their matched type (service delay 1) and slow otherwise (delay 3). A specialist serves a task at its own site and returns to that site’s pool. The decision is which queued task each free worker takes. Because every worker cycles within its own site, each worker’s token lineage links only decisions that used it, and the episode decomposes into eight causal components—which is why this environment measures $K = 8.0$ and a causal fan-out of exactly 1.

The environment also carries a coupling knob we do not exercise in the main results but which motivates the design: a pool of *flexible* generalists, usable at any site (delay 2, type-independent). A flexible worker that serves site i , returns to the shared pool and then serves site j links those sites’ decisions into one component. The local/flexible capacity split therefore traces the whole independence spectrum—all-dedicated gives n_{sites} components, all-pooled gives one—instantiating the classical OR question of resource pooling and flexibility as a single dial over K . The reported configuration is the all-dedicated end.

The N -copies scaling benchmark. N fully independent single-stage assignment problems in one net, served by one shared policy, with reward summed over copies. Each copy has its own arrival stream, queue, busy place and dedicated employee; no place is shared. A copy’s tasks are either *matched* to its employee (service $1 + U\{0, 1\}$) or *crossed* (service $3 + U\{0, 1\}$), so serving matched work first maximizes throughput. The copies are causally independent by construction, so $K \leq N$ with equality whenever every copy emits a reward within the horizon (Definition 1). This is the one environment where the independence structure is a controlled parameter rather than a measured property, which is what makes it the scaling benchmark; it is deliberately synthetic.

The hard N -copies variant. The same net with a genuine per-copy assignment. Each copy has two specialists (codes 0 and 1) and tasks of three types, arriving one per unit time with type uniform on $\{0, 1, 2\}$ and a warm start of one task of each type. Specialist k serves type k in delay 1; type 2 matches neither specialist and takes 3; a cross-assignment (type 0 to specialist 1 or the reverse) takes 4; each plus $U\{0, 1\}$. The decision is therefore which queued task each free specialist takes, and the matched-first rule that anchors the normalization is not optimal, since it does not prefer a type-2 task to a cross-assignment nor choose between the two servers. No place is shared across copies, so the independence structure is unchanged. Section 6.7 reports it at $N = 4$; the anchors are 47.55 (random) and 75.25 (heuristic).

The single-component environment (negative control). A two-stage service line: cases arrive stochastically, are served at stage one, then queue for stage two, and reward is emitted on completing stage two. Both stages draw from *one shared pool of three heterogeneous employees*—two specialists, fast on their matched type (delay 1) and slower otherwise (2), plus a generalist (delay 3)—and every employee returns to that same pool after each service. The shared pool links every case’s lineage into a single causal component, giving $K = 1.0$ with one component holding

Table D.8: Environment parameters. Horizon is in simulated time units; the heuristic is the anchor for the normalization (score – random)/(heuristic – random). The last column is the exact optimum where we can compute it (Section 6.6), expressed on the normalized scale, so it is the ceiling any arm in Table 3 could attain. On N -copies the arms may postpone and the ceiling is the idling-permitted optimum; the anchors never postpone, and against the non-idling optimum the heuristic is exactly optimal there (1.000). On multi-site neither arms nor anchors may postpone, so the ceiling is the non-idling optimum. Dashes are configurations whose optimum we have not computed.

environment	structure	horizon	random	heuristic	ceiling
multi-site routing	4 sites $\times$ 2 dedicated specialists	20	62.90	80.25	1.029
N -copies, $N = 2$	2 independent copies, 1 employee each	20	15.87	22.60	1.010
N -copies, $N = 4$	4 independent copies, 1 employee each	20	31.13	44.60	1.010
N -copies, $N = 4$, hard	4 independent copies, 3 types $\times$ 2 specialists each	20	47.55	75.25	—
N -copies, $N = 8$	8 independent copies, 1 employee each	20	63.20	89.53	1.010
single component	2 stages, 1 shared pool of 3	20	9.85	14.78	—

100% of reward mass. It is the mechanism’s own predicted null, and it is also the environment of the boundary-condition investigation of Appendix G, where it is called the *bottleneck environment* because the shared pool is the binding constraint.

Heuristics.. All three anchors are simple priority rules and none ever postpones. On multi-site, a specialist takes its own matched type first; failing that a flexible worker is used; failing that any specialist takes crossed work. On N -copies, a binding serving a matched (fast) task is preferred, otherwise any enabled binding is taken. On the single-component environment the rule prioritises by stage—work in progress at stage two before new work at stage one—and within a stage prefers a type-matched employee. On the hard N -copies variant *matched* means the task type equals the specialist’s code, and the rule is otherwise the same. Random baselines sample uniformly from the enabled bindings. Both anchors are evaluated on the canonical no-postpone variant of each net, so the normalization scale is a property of the environment rather than of the action space the learner sees.

Appendix E. Training setup and reproducibility

Architecture.. Policy and value networks are separate heterogeneous graph networks over the assignment-graph representation, each a stack of three Heterogeneous Graph Transformer convolutions (Hu et al., 2020) with hidden width 128 and two attention heads, with residual connections and dropout. Places and transitions are distinct node types. The critic pools over node embeddings to produce a scalar; the actor decodes one logit per enabled action binding from that binding’s node embedding—the asymmetry Appendix G analyses. Input widths are inferred lazily from the first observation, so the same code runs unchanged across environments with different attribute sets.

Optimization.. All arms share the hyperparameters of Table E.9; no per-arm tuning was performed, and the causal schemes inherit the baseline’s settings unchanged. This is deliberate—any tuning advantage would confound the

Table E.9: Hyperparameters, identical for every arm; the two rows that differ by environment say so.

parameter	value
algorithm	PPO with clipped surrogate objective
epochs	40 (converged budget; 15 and 30 retained as ablations)
episodes per epoch	20 (multi-site, single component); 8 (N -copies)
minibatch size	32
policy / value learning rate	$3 \times 10^{-4} / 3 \times 10^{-4}$
policy / value update passes	3 / 4 per epoch
clip parameter ε	0.2
entropy bonus	0.01
KL early-stopping limit	0.15 (mean per-state $\text{KL}(\pi_{\text{old}} \pi)$)
discount γ / GAE λ	0.99 / 0.95
SMDP discount rate β	0.5
postponement	enabled (N -copies, single component); disabled (multi-site)

credit comparison—but it also means the reported numbers are not the best attainable for any arm. For the baseline alone we checked that this does not explain its multi-site result: a grid of one-knob departures from the shared setting (policy and value learning rate 10^{-4} and 10^{-3} , $\lambda = 0.9$, entropy bonus 0.03), each on the same 20 seeds at 40 epochs of eight episodes, gives final normalized returns of 0.15–0.28 with 9–14 collapsed seeds per configuration, none better than the shared setting itself. The grid was run at the smaller per-epoch budget, so its absolute values sit below the 20-episode multi-site column of Table 3; it is the ordering within the grid that answers the tuning question.

Evaluation protocol. Every second epoch (every third on N -copies) the greedy (argmax) policy is evaluated on 20 episodes; the reported *final* value is the last such point and *best* the maximum over them. All evaluation uses common random numbers: with the evaluation seed fixed, evaluation episode i always runs scenario seed + i , so every method, seed and epoch is scored on one fixed scenario set and arms differ only by policy. This matters more than it may appear: without it, the measured noise floor on the single-component environment is ± 0.231 standard deviations per evaluation point and ± 0.327 on a paired difference, and drift is inflated by roughly 0.40 even for a policy that is not changing. Baselines average 40 episodes. Seeds 0–19 are used for every cell, and comparisons are paired seed-by-seed; we report paired t -tests and Wilcoxon signed-rank tests, and where they disagree we say so.

Determinism. A single entry point seeds Python’s random, NumPy and PyTorch, and is applied to environment construction, network initialization, rollout collection and evaluation, so a cell is reproducible from its (environment, method, seed) triple. Paired arms share their network initialization. Each cell is written to its own result file and runs are resumable: re-running a sweep trains only the missing cells, which is how the arm coverage in Table 3 was filled incrementally.

Cost. Credit computation is a single backward pass over the decisions of one episode, $O(|E|)$ in the provenance DAG, measured at approximately one millisecond per episode. Against a training cell at four copies this is under 8% of wall-clock time (PPO 7.44 minutes against 8.05); the graph network dominates, and profiling attributes roughly 83% of a cell to the optimizer’s repeated forward and backward passes over the collected graphs. No arm requires forked simulation.

Appendix F. Falsified mechanisms

Section 6.5 reports that the arm computing the causal estimand exactly is not the best performer, and offers only a partial structural account of why the remaining variants order as they do. Rather than advance a mechanism we cannot support, we record every candidate we tested and rejected, with the measurement that rejected it, so that a reader can judge how much of the ordering is explained and how much is not. All six were pre-registered in the sense that the prediction was written into the diagnostic script before it was run.

Appendix F.1. Summary

Appendix F.2. Detail

Postpone transparency. Under token-flow postponement a `POSTPONE` firing re-emits the entire marking, so it terminates the nearest-producer walk without having caused the flow, and it carries $2.25\text{--}2.66\times$ the row sum of production actions. We therefore expected removing it to make R closer to action-measurable. It does the opposite. Transparency *redirects* postpone’s edges to the true producers—0 production-to-production edges lost, 50 gained at four copies and 97 on the single-component environment—and the recovered edges are precisely the action-dependent ones. Postpone had been making R look invariant by discarding information. The resulting variant is also the worst arm we measured (0.106 normalized), so on this environment the more provenance-faithful graph is the weaker learner—a second instance of the pattern in Section 6.

Chain decay and credit magnitude. Both were attempts to explain a large performance gap by a mechanical property of the credit vector, and both fail on magnitude grounds: the measured differences are one to five percent where the performance difference to be explained was 0.44 in normalized return.

Ranking fidelity. This is the sharpest negative result, because it tests the estimator against ground truth rather than against another estimator. At a decision state we fork the simulator once per action under common random numbers and record the realized discounted return-to-go; differences across branches are then the causal effect of the action with the exogenous component cancelled. A credit scheme should rank actions the way that quantity does. On the single-component environment `CGAE-DAG` ranks them *better* than `CGAE-CF` on both measures while learning far worse. Ranking fidelity at a fixed state therefore does not determine learning outcome. Caveats: 12 usable fork points, and the continuation policy is random, so this measures counterfactual value under random continuation rather than under the converged policy.

Table F.10: Candidate mechanisms for the ablation ordering, and the measurements that falsified them. “Inverted” means the measured quantity ordered the variants *opposite* to performance.

hypothesis	measurement	outcome
Postpone is the dominant source of the action-dependence of R	making postpone transparent leaves R identical across actions on 50.0% of fork points, down from 68.8%; spread on the single-component environment rises $0.580 \rightarrow 0.947$	worse
Credit decays along chains as $\prod w$, so CGAE-F propagates less	backward propagation mass at causal depth 5: 0.89 (CGAE-F) vs 0.95 (CGAE-CF)	too small
Larger credit magnitude gives a larger effective step	mean $ Q $ at two copies: 2.23 vs 2.22; SD 1.10 vs 1.12	identical
Per-state ranking fidelity against a measured counterfactual	Spearman ρ / top-1 agreement with common-random-number counterfactual return: CGAE-DAG +0.681 / 83.3%, CGAE-CF +0.435 / 33.3%	inverted
Credit signal-to-noise ratio (bias–variance)	between-action signal over within-action noise: CGAE-DAG 0.959 (highest), CGAE-CF 0.895; full range 0.80–0.96	inverted
CGAE-DAG’s arbitrary $\lambda^{\min}$ depth weighting	replacing it with the depth-conditioned weight $\mathbb{E}[\lambda^L \mid \text{visit}]$ moves correlation with CGAE-CF from 0.740 to 0.748	$\Delta = 0.008$

Signal-to-noise.. We forked each production action under M shared exogenous seeds and formed the ratio of between-action to within-action credit dispersion—an F -ratio for the quantity a policy gradient integrates. The prediction was that CGAE-DAG, whose closure sum approaches the unscoped return on a single-component problem, would show the lowest ratio. It shows the highest. The mechanism was half right and immaterial: CGAE-DAG does carry about twice the noise (0.459 vs 0.272), but it also carries about twice the signal (0.420 vs 0.237), and the scaling cancels.

The depth-weighting choice.. CGAE-DAG weights a descendant’s owned reward by λ^k at minimum causal depth k , which is discretionary: a re-convergent descendant reachable at depths 2 and 7 receives λ^2 . We built the principled alternative, weighting by $\mathbb{E}[\lambda^L \mid \text{the walk visits } j]$, which coincides with the closure estimand at $\lambda = 1$ and inherits CGAE-CF’s depth distribution conditioned on arrival. It changes almost nothing. The reason is structural and is worth stating as a positive by-product: since

$$c_d^{\text{CGAE-CF}}(j) = \underbrace{\Pr[\text{visit } j]}_{\text{the fan-out gap}} \cdot \underbrace{\mathbb{E}[\lambda^L \mid \text{visit } j]}_{\text{depth weighting}},$$

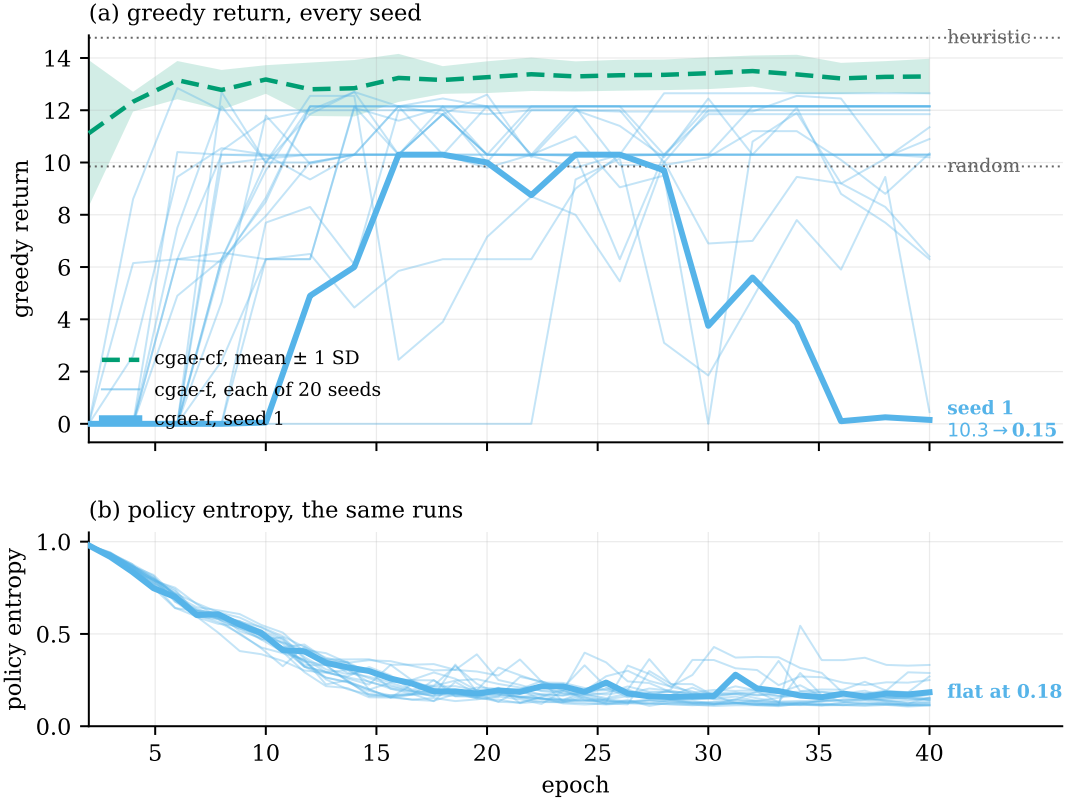


Figure F.7: Per-seed behaviour of CGAE-F on the single-component environment. **(a)** Every CGAE-F seed (thin) against CGAE-CF’s mean and spread (dashed, band); 10 of 20 CGAE-F seeds end collapsed. The emphasised run is seed 1: it climbs to 10.3, holds there for roughly six evaluation points, then falls to 0.15—below the random anchor. **(b)** Entropy over the same runs, on a shared epoch axis. Seed 1’s entropy is flat at 0.18 throughout the collapse, so the end state is a low-entropy poor policy rather than a failure to explore, which is the pattern Proposition 2 leads one to expect when the error term grows with the critic. A mean with a confidence band does not show this, which is why the behaviour was not apparent in five-seed comparisons.

the two factors are separable, and the closure variants differ from CGAE-CF only by discarding the first. The behavioural difference is the gap itself, not any detail of how depth is discounted within the closure.

Appendix F.3. Best-checkpoint reading of the ablation ladder

Table 4 reports final greedy return. Under best-checkpoint restoration the two significant results and the whole ordering are unchanged: CGAE-F reads 0.443_(0.14) (+0.390 against CGAE-CF, 20W/0L, $p < 0.0001$) and CGAE-DAG 0.516_(0.15) (+0.316, 19W/0L, $p < 0.0001$), against CGAE-CF’s 0.833_(0.08), with every remaining arm within 0.041 of CGAE-CF and no p below 0.13. The magnitudes shrink because checkpoint restoration is a partial guard against the late-training deterioration of Section 6.4.

Appendix F.4. A budget artifact, recorded for the same reason

At the 15-epoch budget with 20 seeds, cGAE-CF measured 0.401 against cGAE-F 's 0.837 at two copies (4W/14L, $p = 0.0043$)—an apparently significant deficiency in the method we ship. It is an artifact of non-convergence. At 40 epochs the comparison is +0.102 ($p = 0.51$), sign reversed. Three measurements identify the cause: a λ sweep recovers most of the deficit ($0.401 \rightarrow 0.579 \rightarrow 0.639$ as λ falls $0.95 \rightarrow 0.85 \rightarrow 0.70$ toward cGAE-F 's effective $0.95 \times 0.896 \approx 0.851$), so it is a convergence-*speed* effect; the credit vectors differ by about one percent; and the outcomes are bimodal with nothing in between. We report this because it is the clearest evidence in this work that a significant result at a short budget can be entirely spurious, and it is why Section 6 reports the converged budget throughout.

Appendix F.5. The one surviving correlate

Across all arms and environments the single variable that separates working from failing configurations without overlap is *final policy entropy*: 0.27–0.56 for the four arms that learn, 0.13–0.15 for the three that fail. Whether this is cause or symptom is untested—a policy in a poor local optimum will show low entropy regardless of how it arrived—and the decisive experiment, an entropy-bonus intervention on the failing arms, has not been run. We state the correlation and its status rather than promoting it to a mechanism.

Appendix F.6. Methodological note

All six diagnostics share a design feature that may explain their uniform failure: each measures a *static* property of the estimator under a *random* policy with random continuations. Training involves a learned policy that becomes progressively deterministic, and the failures we are trying to explain unfold over tens of epochs. It is entirely possible that no static, random-policy property of a credit estimator predicts its training outcome. If so, that is itself a finding about how credit-assignment methods should be evaluated, and it argues for interventional rather than observational diagnostics—consistent with the boundary-condition results of Appendix G, where only active-intervention mechanisms ever moved the single-component floor.

Appendix G. The boundary-condition investigation

Proposition 3 predicts that cGAE-CF 's variance reduction collapses to exactly PPO when $K = 1$. That is the proposition's own null, and it raises the question Section 7.5 summarizes: whether the residual gap on a genuinely single-component problem can be closed by some other provenance- or structure-based route. We tested three independent candidates, none of which touches credit assignment; all three change only what the learner observes. Each is provably inert or empirically neutral, and two of the three results follow from a proof rather than from a correlation, which is why we report them rather than omit them: together they indicate where the difficulty on this environment does not lie. The environment is the two-stage service line with one shared, skill-heterogeneous employee pool, called *single component* in Section 6 and the *bottleneck environment* here. Table G.11 summarizes the three outcomes.

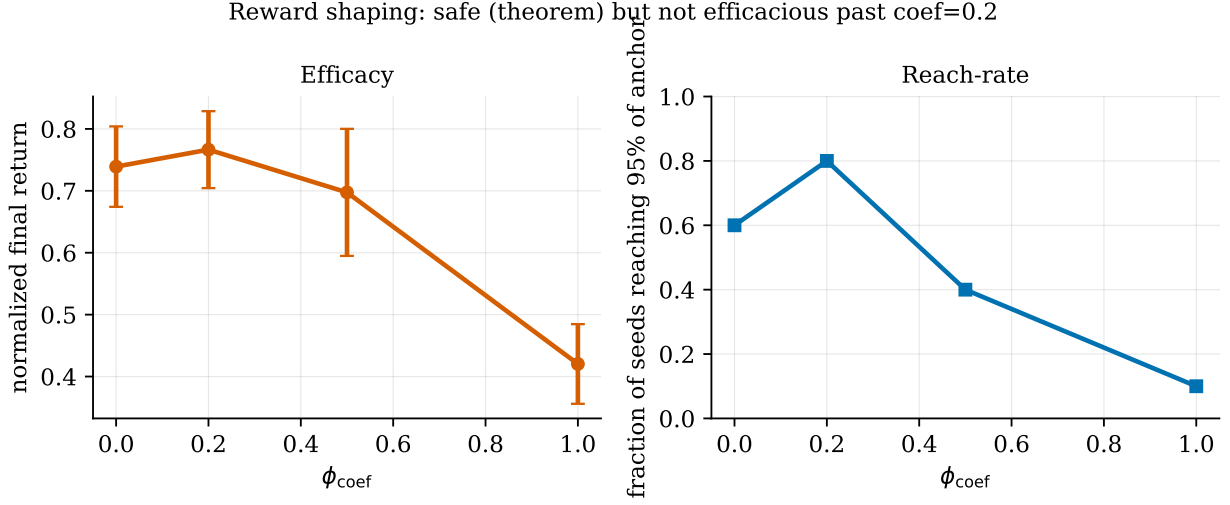


Figure F.8: The reward-shaping coefficient dial on the bottleneck environment: efficacy (left) degrades once past $\phi_{\text{coef}} = 0.2$, and the fraction of seeds reaching 95% of the heuristic anchor (right) degrades monotonically—safety (Section 2, Ng et al., 1999) and efficacy are independent properties, and this mechanism has the first without the second at practical training budgets.

Table G.11: The boundary-condition investigation, bottleneck environment ($K=1$).

mechanism	theoretical status	result
Reward shaping	policy-invariant, any potential	neutral (0.2) to harmful (1.0), 0W/10L
Structural input features	mechanically correct	neutral, $p = .58 \rightarrow .69$; provably redundant
Actor global context	proven to close a real gap	neutral, $p = .91$; gap was already marginal here

Protocol, and why it differs.. The three experiments of Appendix G predate the converged-budget re-runs and use the stochastic suite’s standard budget: the single-component environment at $30 \text{ epochs} \times 20 \text{ episodes}$, 10 seeds, paired seed-by-seed against a `ppo_clip` baseline, reporting normalized final return. That baseline therefore reads 0.739 ± 0.10 here against 0.729 ± 0.12 in Table 4, which is the same arm at 40 epochs and 20 seeds; the difference is budget and seed count, not environment. We did not re-run them at the longer budget because all three outcomes are nulls or negatives, and a longer budget cannot convert a null into the positive result that would change the section’s conclusion. The environment is the one described in Appendix D; we call it the *bottleneck environment* to name the mechanism under test rather than the statistic K .

Potential-based shaping.. The shaping term is the SMDP generalization $F(s, a, s') = e^{-\beta\tau} \Phi(s') - \Phi(s)$, added to every step’s reward, with the potential read off the net’s topology: $\Phi(s) = \sum_p \text{decay}^{\text{hop}(p)} \cdot \min(|p|, \text{cap})$, a distance-decayed structural backlog in which each place’s token count is weighted by its hop-distance to the nearest reward-emitting transition and optionally capped. Policy invariance (Ng et al., 1999) was verified in code by an exact terminal-value cancellation test rather than merely cited. The coefficient dial, shown in Figure F.8, reuses one cached baseline across

all four points, so the comparison is paired by construction.

The failure mechanism is worth stating because it is not a property of the potential’s shape. This environment has constant exogenous arrivals—one case per time unit—so the backlog potential rises as often as it falls, and the per-step shaping signal is of the same order of magnitude as the entire episode return. Aggregated over an episode the term telescopes to a constant, exactly as the theorem promises, but at any finite budget it is a substantially noisier value-regression target. Safety and efficacy are independent properties, and this mechanism has the first without the second.

Structural input features.. Two feature sets were appended to each action node’s own inputs: first, whether an action type structurally competes for a token and with how many rivals, derived from the conflict graph; then, after that returned a null, the minimum topological hop-distance from an action’s required inputs to the nearest reward-emitting transition. The second does discriminate between the environment’s two action types (0.656 against 0.810), so the null is not a degenerate-feature artifact.

The redundancy argument is the stronger result and does not depend on either feature’s design. Every action node already carries a one-hot action-type identity at zero hops. Any scalar that is a function of action type alone is therefore a deterministic function of information a single learned weight on that one-hot could reconstruct, so it cannot add representational capacity. This holds for this architecture on any topology, which is why we report it as a proof rather than as two failed experiments. A structural feature can only help if it varies *within* an action type—across markings—which conflict-degree on a two-node conflict pair provably does not.

Actor architecture.. The blind spot is a property of the computational graph, not of training. The actor decodes each action’s logit from that action node’s own final message-passing embedding; the graph’s edges run strictly along token flow, with no reverse edges at any real observation call site; so an action’s logit can depend on state only through a forward path reaching it within the network’s depth. The demonstration uses a purpose-built topology of two structurally independent two-stage pipelines: perturbing one pipeline’s downstream marking leaves the other pipeline’s action logit bit-for-bit unchanged ($\Delta = 0.000$) under a randomly initialized, untrained network. Reachability does not depend on learned weights, so no training budget can close it.

The fix concatenates onto each action node, before decoding, the same pooled context the critic already computes—an intervention on the actor alone, leaving the critic and the credit path untouched. On the proof topology it closes the gap exactly ($\Delta = 0.000 \rightarrow 0.528$). On the single-component environment it is neutral (0.746 ± 0.14 against 0.739 ± 0.10 , $p = .91$), and the two results are not in tension: that environment’s shared resource pool recycles workers, which gives the unfixed architecture a marginal, accidental path to the same information within three hops. The blind spot was never total there, and whatever fraction the fix newly exposes is not decision-relevant enough to move a well-tuned baseline at this budget. The architectural finding stands on the proof; the training result bounds its practical value on this environment only.

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